



GS FOUNDATION 2024-25

GEOGRAPHY - 14

Distribution of Oceans and Continents - I

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Types of Mountain

- ① Block Mountain ② Fold Mtn. ③ Residual Mountain
④

① Fold Mountain

- Very important to understand earth crust.
- Marine sediments & have long linear chain of mountain composed of
- Fold mountain form tallest mountain chain. formed due to folding of sediment due to compressive forces.

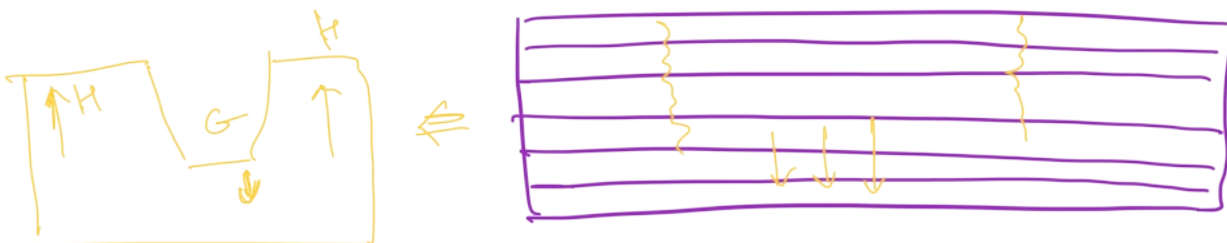
Eg. Alps, Rockies, Andes, Himalaya, Eastern Ghats Aravali

Block Mountain

- Formed due to faulting
 - ↳ cracking → creation of fracture
 - ↳ drifting
 - ↳ displacement up & down

→ Formed when gigantic area are displaced vertically. uplifted block are called HORST.

→ & low line are called GRABEN.



It happen due Tensile force { Tension }
OR
compressive force

Fault formed due to tensile force are called NORMAL FAULT.

" " compressive force " " REVERSE FAULT

Es. VOSGES, Western ghat, Sierra-Nevada
HARZ

VOLCANIC MOUNTAIN :-

formed due to volcanic Activity

→ Mountain Fujiyama, Some island of Andaman & Nicobar

Residual Mountain

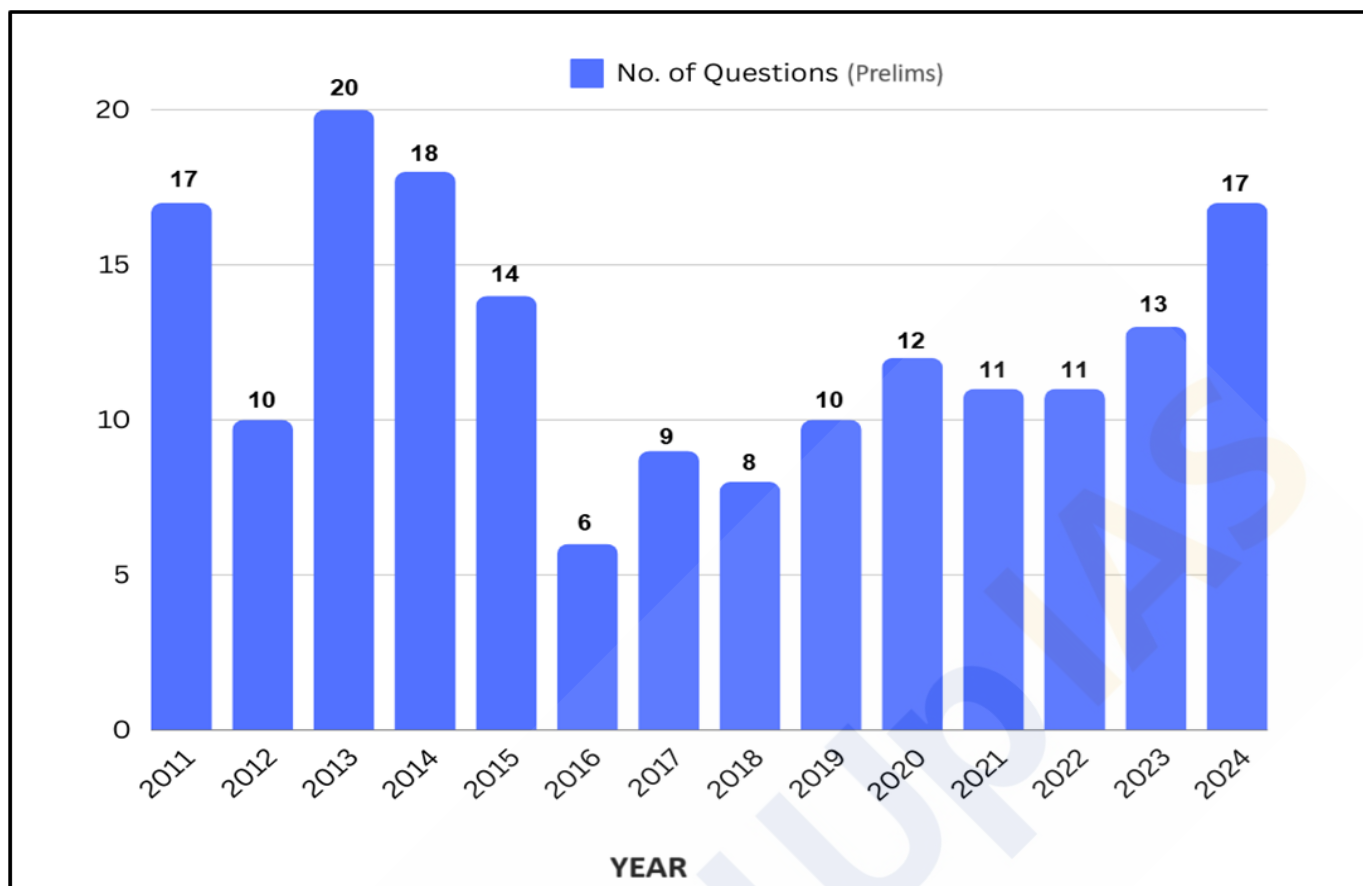
Residual of old Fold mountain when it gets eroded.
OR

When block mountain get eroded.

Eg. Vidhalhal

Not growing anymore

Year-wise Trend Analysis (2011 – 2024)



Topic-wise Trend Analysis (2011 – 2024)

Physical Geography

Topic	Questions
Geomorphology	9
Oceanography	5
Climatology	22
Mapping	16
Distribution of Resources	5
Miscellaneous	15

Indian Geography

Topic	Questions
Physiography of India	24
Drainage System of India	15
Indian Climate	7
Indian Soils	5
Natural Vegetation of India	12
Indian Agriculture	31
Mineral Resources of India	10

Distribution of Oceans and Continents

Introduction:

Continents and ocean basins, as fundamental relief features of the globe, are considered "**relief features of the first order.**" Therefore, it is crucial to investigate their possible origins and evolution. Over time, scientists have proposed various views, concepts, hypotheses, and theories regarding the origin of continents and ocean basins. Before delving into these theories, it is essential to understand the current distributional patterns and arrangements of continents and ocean basins.

Approximately 70.8% of the Earth's surface is covered by oceans, while the remaining 29.2% is represented by continents. The distribution of continents and oceans is **not uniform across the hemispheres.**

Land and Water Distribution:

Dominance of Land in the Northern Hemisphere:

- **More than 75% of the Earth's total land area is located north of the equator,** making the northern hemisphere predominantly land. In contrast, the **southern hemisphere is dominated by water bodies.** Dividing the globe into hemispheres with the north pole in the English Channel and the south pole near New Zealand, the northern hemisphere becomes the "**land hemisphere**" (83% of the total land area), while the southern hemisphere is the "**water hemisphere**" (90.6% of the total oceanic area).

Triangular Arrangement of Continents:

- **Most continents have a roughly triangular shape, with their bases in the north and their apices pointing south.** For instance, North and South America together form an equilateral triangle with the base along the Arctic Sea and the apex at Cape Horn. Separately, these continents also form individual triangles. Similarly, Eurasia forms a triangle with its base along the Arctic Sea and its apex near the East Indies. Africa's base is towards the north, with its apex at the Cape of Good Hope. **Australia and Antarctica are exceptions to this rule.**

Triangular Shape of Oceans:

- Contrary to continents, **oceans are also triangular but with their bases in the south and apices in the north.** The Atlantic Ocean's base stretches between Cape Horn and the Cape of Good Hope, with its apex east of Greenland. The Indian Ocean's base is in the south, with apices in the Bay of Bengal and the Arabian Sea. The Pacific Ocean's apex is near the Aleutian Islands, and its base lies in the south.

Polar Surroundings:

- The **north pole is surrounded by oceanic water,** while the **south pole is surrounded by the land area of Antarctica.**

The Pacific Ocean Basin

- The **Pacific Ocean** basin occupies almost **one-third of the Earth's entire surface area.**

Theories on the Origin of Continents and Ocean Basins:

Less than 420 years ago, many **scientists thought the continents always had been in the same place.** It was difficult to accept that the continents and oceans are moving across the surface of the Earth although the idea about united continent was not new. A few scholars used to wonder that South America and Africa appear to fit together. For example, **Abraham Ortelius** in 1596, **Sir Francis Bacon** in 1620 and **Antonio Snider Pellegrini** in 1858 in his book '**Creation and Its Mysteries Unveiled**', noted that the **eastern coastline of South America and the western coastline of Africa** looked as if they could fit together.

Alexander von Humboldt one of the founder of modern Geography during his scientific expedition (1799-1804) across the Atlantic further grasped that this similarity is more than the apparent “fit”. The mountains of Brazil in South America were the same as those of the Congo in Africa. Therefore, **similarity between eastern coastline of South America and the western coastline of Africa is not accidental.**

However, it was **Alfred Lothar Wegener 1912** – a 32-year-old lecturer in meteorology and astronomy in Germany who propounded the ‘**Continental Drift Theory**’ with sound evidence

The Continental Drift Theory:

Aim of the Theory:

Wegener introduced the concept of continental drift in **1912**. However, it gained significant attention only in **1922** when he detailed his theory in a book titled “**Die Entstehung der Kontinente und Ozeane**,” translated into English “**The Origin of Continents and Oceans**” in **1924**.

The primary issue Wegener sought to address was the **explanation of climatic changes throughout Earth's history**. He noted substantial evidence of significant climatic shifts over time. Wegener's continental drift theory emerged from the **need to explain these major climatic variations**.

The Climatic changes observed on Earth could be explained in two ways:

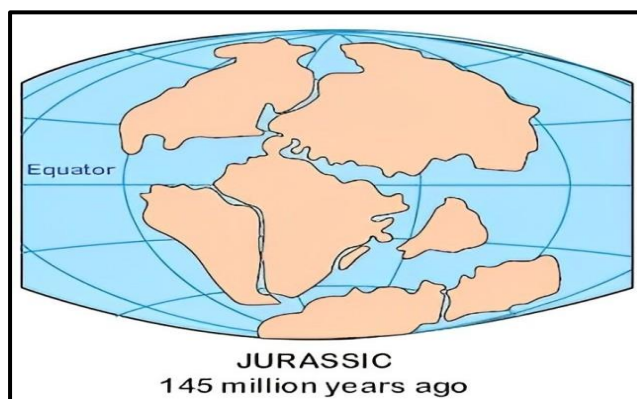
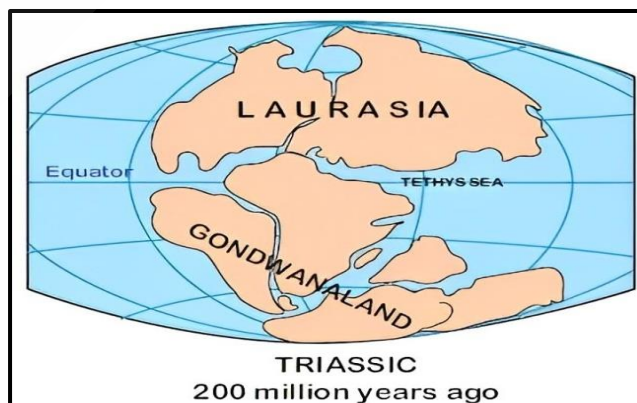
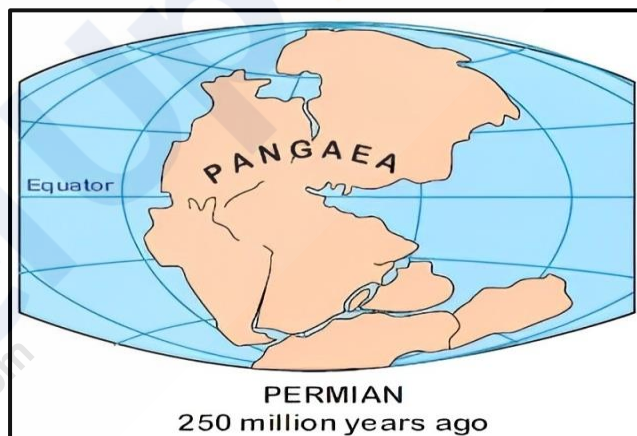
1. If continents remained stationary throughout geological history, climatic zones might have shifted, causing regions to experience different climatic conditions over time.
2. If climatic zones remained stationary, landmasses might have moved and drifted.

Wegener chose the second explanation, rejecting the notion of permanent continents and ocean basins. Therefore, his main objective behind the displacement hypothesis was to explain the global climatic changes reported throughout Earth's history.

Basic Premise of the Theory:

Following Edward Suess, Wegener believed in a **three-layer system of the Earth**, consisting of the **outer layer** of 'sial,' the **intermediate layer** of 'sima,' and the **lower layer** of 'nife.' According to Wegener, 'sial' was limited to continental masses, while the **ocean crust was represented by the upper part of 'sima.'** He proposed that **continents, or sialic masses, floated on 'sima' without resistance.**

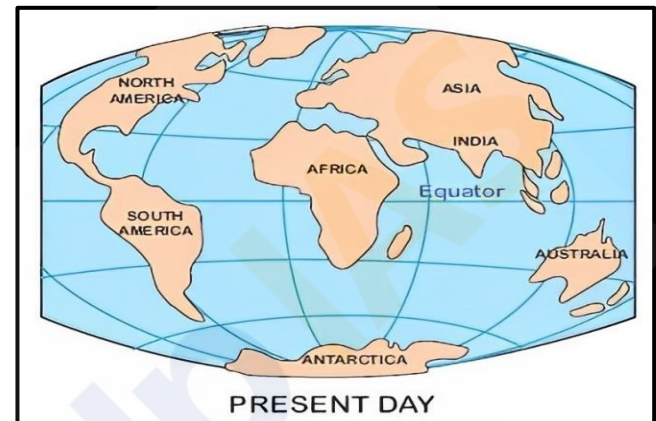
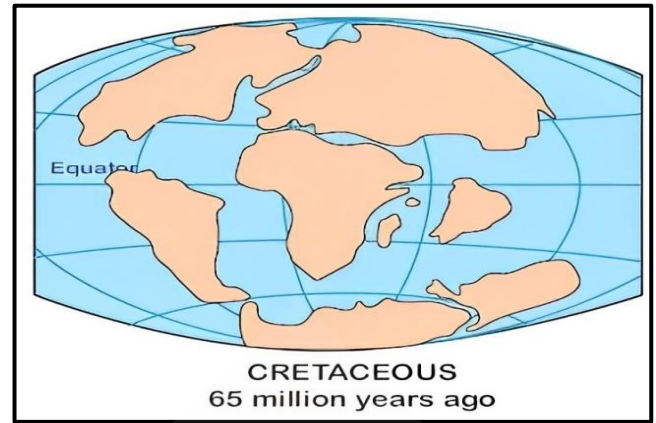
Based on evidence from paleoclimatology, paleontology, paleobotany, geology, and geophysics, Wegener hypothesized that **all landmasses were once united into a supercontinent** called ‘**Pangaea**’ during the Carboniferous period. Pangaea was **surrounded by a vast ocean named 'Panthalassa,'** representing the primordial Pacific Ocean. The **northern part of Pangaea**, called ‘**Laurasia**,’ included present-day North America, Europe,



and Asia, while the **southern part, 'Gondwanaland,'** comprised South America, Africa, Madagascar, Peninsular India, Australia, and Antarctica.

He proposed that, around **200 million years ago**, during the **Mesozoic era**, **Pangaea began to separate into two proto-continents** - northern continent (**Laurasia**) and southern continent (**Gondwanaland**). The **Tethys Sea progressively filled the area between Laurasia and Gondwanaland** during the Mesozoic epoch, and it gradually broadened. The continents drifted away from the contemporary poles. Wegener described this phenomenon as **Polflucht (flight from the poles)**.

By the **end of the Cretaceous period**, the continents were **separating into land masses** that resembled the shape of our **present-day continents**. Wherever the drifting continent faced resistance in the way of movement it resulted in huge fold mountains. Therefore, his book provided a new reasoning for the construction of mountain chains.



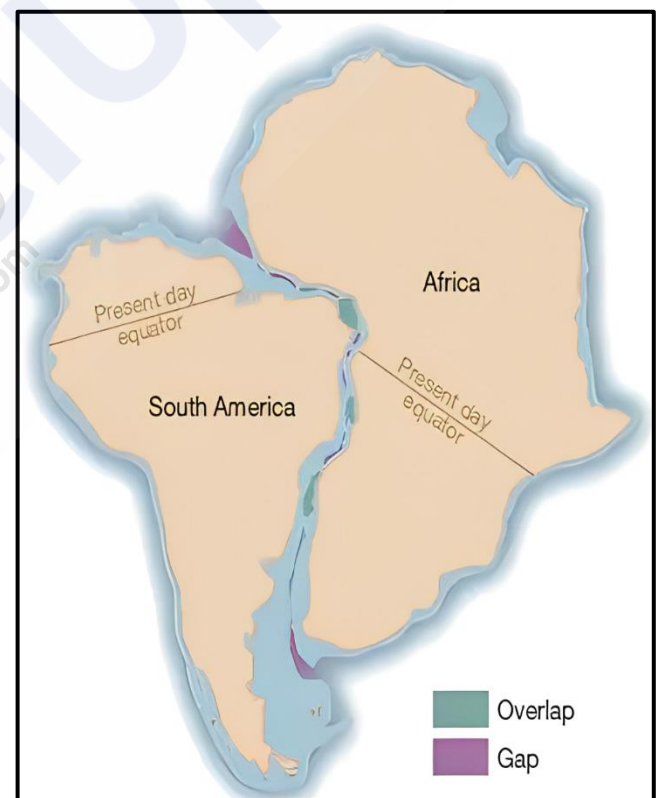
Evidence in Support of the Theory:

The Matching of Continents (Jig-Saw-Fit):

- One of the compelling pieces of evidence for the Continental Drift Theory is the remarkable **fit between the shorelines of Africa and South America**. When these continents are aligned, their coastlines **match with striking precision**. This phenomenon was further substantiated in 1964 when Bullard used a computer program to map the Atlantic margins, revealing an almost perfect fit at the 1,000-fathom line instead of the current shorelines.

Rocks of the Same Age Across Oceans:

- Radiometric dating** has enabled geologists to **correlate rock formations across vast oceans, revealing that ancient rock belts on different continents are of similar age**. For instance, a belt of ancient rocks dating back 2,000 million years on the **coast of Brazil matches those found in western Africa**. Furthermore, the earliest marine deposits along the coastlines of South America and Africa are from the Jurassic age, indicating that the Atlantic Ocean did not exist before this period.



Tillite:

- Tillite, a sedimentary rock formed from glacial deposits**, provides significant evidence for continental drift. The **Gondwana system of sediments found in India has counterparts in six different landmasses in the Southern Hemisphere**. These systems contain thick tillite layers at their base, indicating extensive and prolonged glaciation.

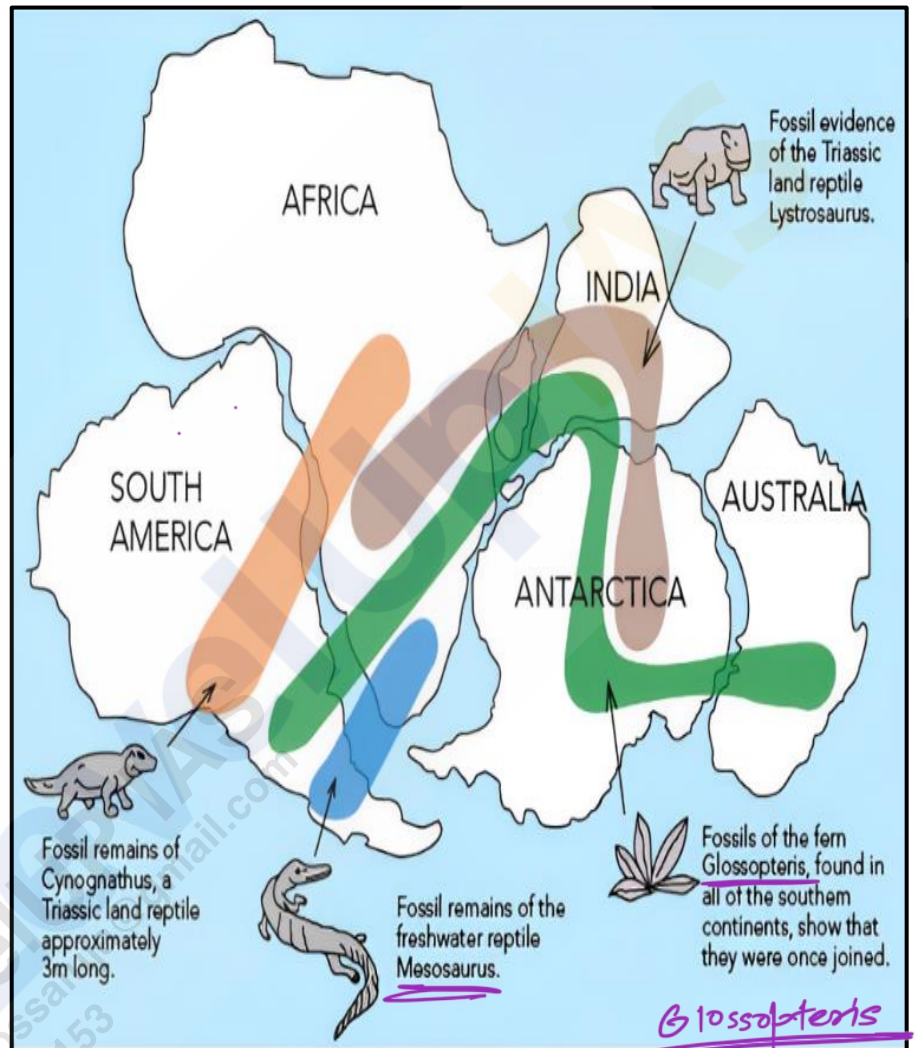
Similar tillite deposits are found in Africa, the Falkland Islands, Madagascar, Antarctica, and Australia, demonstrating that these landmasses shared a common glacial history and further supporting the idea of continental drift.

Placer Deposits: (Gold deposit)

- The occurrence of rich placer gold deposits on the Ghana coast, without any nearby source rocks, suggests continental drift. The gold-bearing veins are found in Brazil, implying that the **gold deposits in Ghana originated from the Brazilian plateau** when the two continents were adjacent.

Distribution of Fossils:

- The distribution of **identical species of plants and animals on continents now separated by oceans** presents another problem that continental drift helps solve. For example, the **presence of lemurs in India, Madagascar, and Africa** led to the hypothesis of a now-submerged landmass, Lemuria, that once connected these regions. Similarly, the **small reptile Mesosaurus**, adapted to shallow brackish water, is found in only two locations: the Southern Cape province of South Africa and the Iruver formations of Brazil. These sites are now 4,800 km apart, separated by an ocean, indicating that these continents were once connected.



Polar Wandering:

- Evidence of polar wandering, which involves **periodic changes in the position of Earth's magnetic poles** recorded in rocks, supports the Continental Drift Theory. Minerals like magnetite and hematite retain permanent magnetism aligned with Earth's magnetic field at the time of their formation. **Geological studies have traced these changes, revealing the shifting positions of the continents over time.**

Paleoclimatic Evidence (old glacier evidence)

- Carboniferous Glaciation** - Glacial deposits from the Carboniferous period found in regions such as Brazil, the Falkland Islands, South Africa, Peninsular India, Australia, and Antarctica indicate these landmasses were once part of a single large landmass. The widespread distribution of these glacial deposits suggests that the continents were connected and experienced similar climatic conditions.
- Coal Deposits** - Coal deposits found in temperate and tropical regions of North America, Europe, and Asia indicate that these areas were once situated closer to the equator, supporting the idea that the continents have drifted over time.

Forces Responsible for Continental Drift:

According to Wegener, the forces responsible for this drift operated in two primary directions: **Equatorward** and **Westward**.

Equatorward Drift:

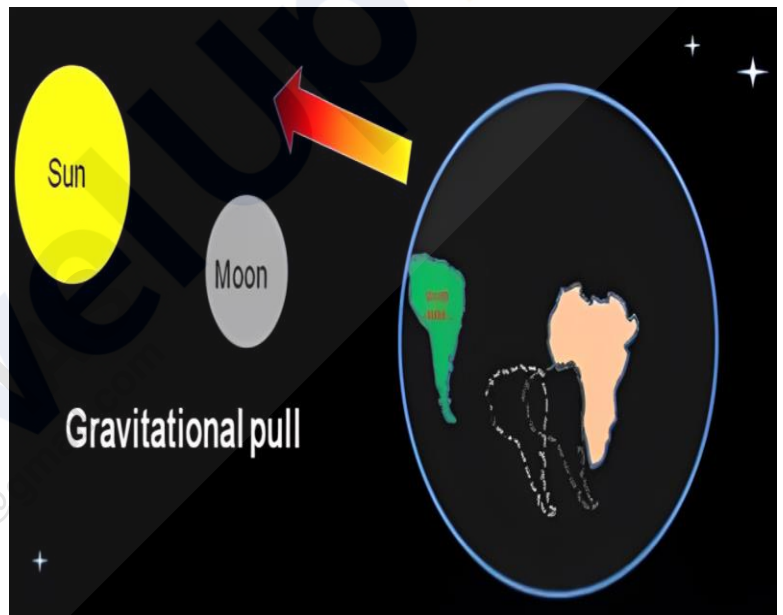
Wegener theorized that continents moved towards the equator due to the combined effects of **pole-fleeing forces** and **gravitational forces**.

- **Pole-Fleeing Force** - The Earth is not a perfect sphere. It has an **equatorial bulge due to its rotation**. The centrifugal force resulting from the Earth's rotation is greater at the equator than at the poles. **This increase in centrifugal force from the poles towards the equator creates a pole-fleeing force**. According to Wegener, this force contributed significantly to the **movement of continents towards the equator**.
- **Gravitational Forces** - The gravitational pull acts to **balance the center of gravity and the center of buoyancy** of the continental blocks. These **blocks, primarily composed of lighter sialic materials** (rich in silica and aluminium), **float on the denser underlying sima** (silica and magnesium). This buoyant effect, analogous to a **ship floating on water**, is one of the primary reasons for the equatorward drift.

Westward Drift:

In addition to the equatorward movement, Wegener proposed that **continents also drift westward**. This westward drift was attributed to the **tidal force**.

- **The Tidal force** - It is due to the **attraction of the moon and the sun** that develops tides in oceanic waters. Wegener believed that these forces would become effective when applied over many million years. He proposed that the **gravitational pull of the sun and the moon**, which was strongest when the moon was closest to Earth, **exerted a drag on the outer sialic crust** (continental blocks), **pulling them westward** over the interior of the Earth.



Wegener's Explanation of Geological Phenomena:

Volcanism and Earthquakes:

- Wegener's explanation for volcanism and earthquakes is rooted in the **interaction between the Sial** (the upper layer of the Earth's crust, composed mainly of silicate and aluminium) **and the Sima** (the lower layer, which consists of silicate and magnesium).
- According to Wegener, the **Sial is hard, brittle, and lighter than the Sima, allowing it to float over the denser Sima**. This **differential in density** causes the Sial to deform, split, and even **rupture the ocean floor**. These ruptures create pathways for magma to escape from beneath the Earth's crust, leading to volcanic activity on the ocean floor.
- In addition to volcanism, the **scraping of the Sial over the Sima generates significant disturbances within the Earth's crust**. These disturbances **create stress and pressure**, which can result in **seismic activity** and, consequently,

earthquakes. Wegener's theory suggests that these **earthquakes are a direct result of the friction and mechanical stress** caused by the movement of the Sial over the Sima.

Formation of Fold Mountains:

- Wegener's theory also extends to the formation of fold mountains. As the **Sial moves across the ocean floor, it scrapes and accumulates marine sediments along its leading edge.** Over time, these sediments build up and become compressed, **forming fold mountains** at the edges of the continents. This process involves the **bending and folding of sedimentary layers**, resulting in the creation of mountain ranges along continental margins.

Formation of Islands:

- According to Wegener, as the **Sial floats over the Sima, it is subjected to significant drag and friction.** This stress particularly **affects the trailing edge of the Sial**, which may **eventually break off due to the accumulated pressure.** The **breaking off of the trailing edge results in the formation of islands.**
- Wegener used the evidence of island formation to support his theory of **continental drift**. He observed that major continents with islands are predominantly located in the southeastern corners, suggesting that the continents have been moving north and westward over geological time. This evidence reinforced his hypothesis of shifting continents and provided additional support for his theory of continental drift.

Criticism of the Continental Drift Theory:

- Controversial Forces** - One of the main criticisms of the Continental Drift Theory was its reliance on speculative forces to explain the drift of continents. Critics argued that **if the gravitational forces from the moon or sun were strong enough to cause landmasses to drift, they would have significantly affected Earth's rotation.** Such forces could have theoretically stopped Earth's rotation altogether or caused it to become stationary.
 - Furthermore, **to cause significant drift in landmasses, the rotational speeds required would be so extreme that they could potentially eject the atmosphere** and other outer layers into space, which is not feasible.
- Directional Drift Anomalies** - Another point of contention was the theory's assertion that **continents drifted mainly northward and westward.** Critics questioned why the drift appeared to follow these specific directions and whether the theory could account for other potential drift patterns.
- Misconceptions about Lithospheric Floats** - The theory suggested that the Sial (continental crust) floated over the Sima (oceanic crust). However, modern understanding shows that the **lithosphere** (which includes both the Sial and Sima) **floats on the asthenosphere.** This distinction undermines the theory's explanation of the drift mechanism.
- Inconsistencies in Mountain Formation** - The theory proposed that mountain ranges like the Rockies and Andes formed due to friction caused by the movement of the Sima. However, this explanation is considered self-contradictory and fails to align with current geological understanding.
- Unexplained Geological Features** - Finally, the Continental Drift Theory **did not adequately explain the formation of oceanic ridges.** These geological features were not accounted for within the framework of the theory, leading to further scepticism and criticism.
- Lack of Precise Historical Evidence** - The theory struggled with the **lack of concrete evidence regarding the pre-carboniferous period (No explanation on what created Pangaea).** This gap in historical knowledge made it challenging to fully understand and validate the theory's claims about the Earth's past.

Contributions of Continental Drift Theory:

Logically Deduced Theory:

- Wegener's theory was grounded in **logical deduction rather than merely observational data**. He proposed that continents had once been joined together and had drifted apart over time. This was a systematic approach to **explaining the similarities in geological formations and fossil records** across different continents. His logical deductions helped to lay the foundation for understanding continental movement.

Bold Departures from Existing Ideas:

- Prior to Wegener, the prevailing view was that continents and oceans were permanent and static features of the Earth's surface. Wegener's theory **challenged this notion by proposing that continents were in constant motion, breaking away from the idea that these features were immutable**. This bold departure from established beliefs opened up new avenues for research and fundamentally changed the way scientists thought about Earth's geology.

Alternative Explanation for Fold Mountains:

- Wegener offered a revolutionary alternative to the then-accepted theory of fold mountain formation. Instead of attributing the formation of fold mountains solely to the uplifting of marine sediments, Wegener suggested that the **collision and movement of continental plates were responsible for creating mountain ranges**. This idea was significant because it provided a new perspective on how geological features could be formed, even though the exact mechanisms he proposed were not fully accurate.

Pioneering Challenge to the Permanency of Continents and Oceans:

- Wegener was one of the first scientists to challenge the idea that continents and oceans were permanent. His theory proposed that **continents had once been part of a supercontinent (Pangaea) and had drifted to their current positions**. This pioneering challenge to the existing view spurred further research and debate, leading to a deeper exploration of Earth's dynamic processes.

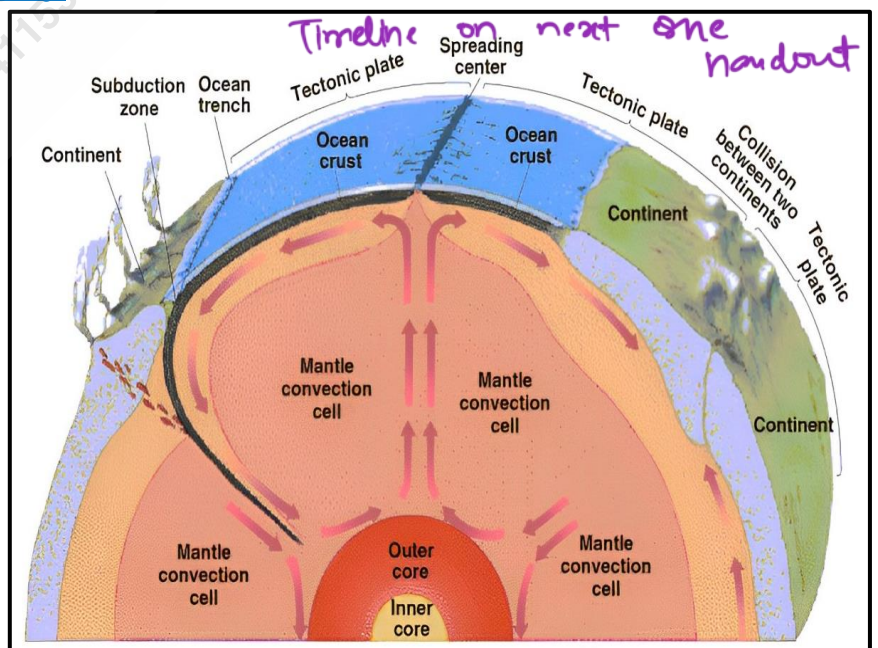
Correct Idea with Incorrect Evidence and Mechanism:

- While Wegener's **basic idea of continental drift was correct, the evidence and mechanism he proposed were flawed**. Nevertheless, his concept was a crucial step in the development of geological theory. The idea of moving continents was later confirmed and expanded upon by the Plate Tectonics Theory, which provided the correct mechanisms (such as seafloor spreading and mantle convection) to explain continental movement. Wegener's work thus played a critical role in the evolution of geological science, despite the initial shortcomings.

Convectional Current Theory *continent moving theory revived by plate tectonic theory*

The Convectional Current Theory, proposed by **Arthur Holmes in the 1930s**, offers a mechanism to explain the **movement of tectonic plates through mantle convection currents**. This theory emerged as a **response to the limitations of the Continental Drift Theory**, particularly in addressing the forces responsible for plate movements.

Convection Current Theory **forms the foundation of the Seafloor Spreading Theory**. According to this theory, the **Convection currents are generated due to thermal differences caused by radioactive elements in the mantle**.

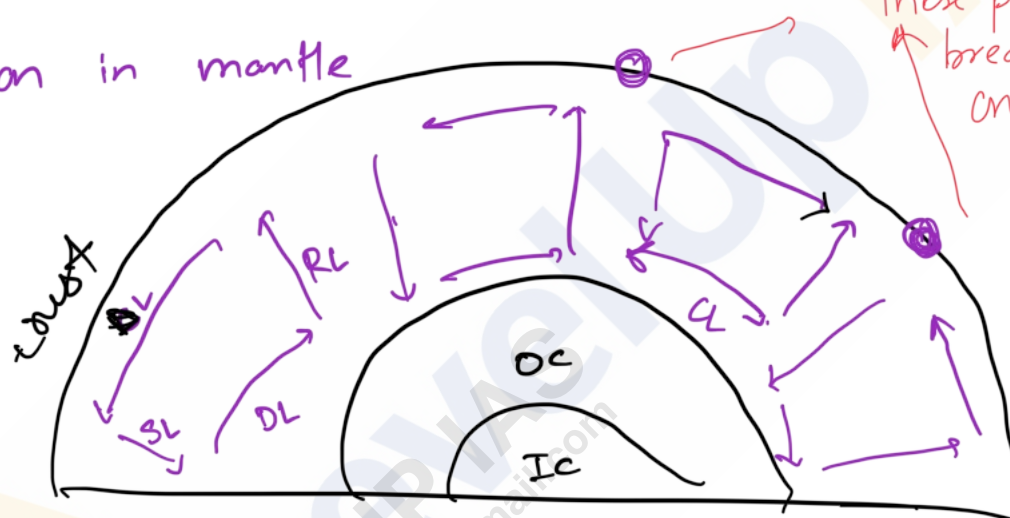


Mechanism:

- **Divergent currents push these blocks apart, leading to the formation of seas and oceans.** Notable examples of this process include the **East African Rift** system and the **Mid-Atlantic Ridge**.
- Conversely, **convergent convection** currents ~~cause the crustal zones to sink, resulting in the formation of geosynclines and the closure of seas~~. This process leads to the creation of **island arcs** and **mountain ranges**.
- Convection currents within the molten rock of the mantle act like a conveyor belt, facilitating the movement of tectonic plates. These currents form closed systems, ensuring that the total amount of lithospheric material remains constant.

Shallow surface on earth that with continuous sink & accumulate sediment. When convergent conv. current, then mountain is formed.

Circulation in mantle



The conv. current are very strong they can broke the lithosphere.

Eg.

Himalaya are formed due to convergence of Eurasian plate with Indian plate.

Paleomagnetism:

Paleomagnetism is the **study of the Earth's magnetic field as recorded in rocks, sediments, and archaeological materials**. By examining the magnetic properties of these materials, scientists can detect changes in the Earth's magnetic field, including its polarity and the occurrence of magnetic field reversals over geological time.

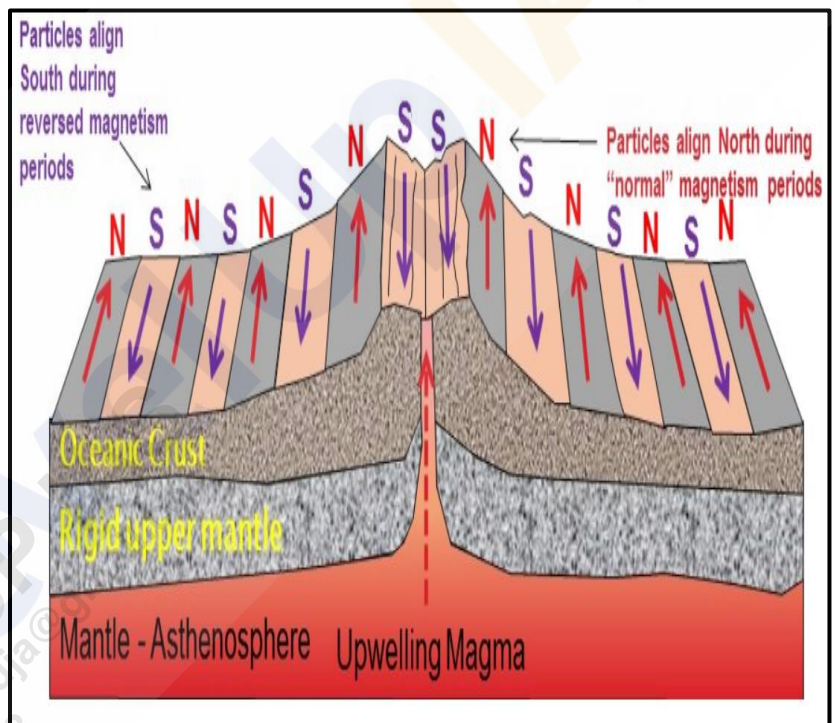
Basaltic rocks, formed from underwater volcanic activity, are primarily composed of **iron-rich minerals** and make up most of the ocean floor. As **basalt solidifies, its magnetic minerals align with the Earth's magnetic field**. This alignment locks in a record of the magnetic field's orientation at the time the rock solidified.

Geomagnetic Reversals:

- Paleomagnetic studies have revealed that the **Earth's magnetic field has undergone numerous reversals throughout geological history**. These discoveries played a crucial role in reviving the continental drift hypothesis and developing the theories of Sea Floor Spreading and Plate Tectonics. One of the key pieces of evidence for these theories comes from **the unique magnetic records preserved along mid-ocean ridges, where the seafloor is actively spreading**.

Magnetic Striping and Sea Floor Spreading:

- At mid-ocean ridges, which are **diverging boundaries** between tectonic plates, **magma rises through fissures or vents and solidifies into new rock**. This newly formed rock records the direction of the Earth's magnetic field at the time of its formation.
- As **tectonic plates move apart due to convective currents**, the solidified rock bands move away from the ridge, and new rock forms in their place. This process creates **alternating magnetic stripes on either side of the ridge, with each stripe represents the orientation of the Earth's magnetic field at the time the rock was formed**. This alternating pattern of magnetic striping provides compelling evidence for Sea Floor Spreading and helps in tracing the past movements and positions of tectonic plates.

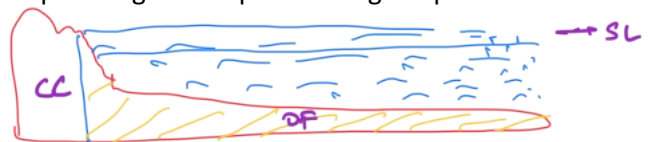


Configuration of the Ocean Floor:

The ocean floor is a complex and varied landscape, much like the surface of the land. It features a range of geological formations including **mountains, basins, plateaus, ridges, canyons, and trenches**. These underwater landforms, collectively known as submarine relief, play a crucial role in shaping the ocean's geography. The ocean floor is generally divided into four main regions - **the continental shelf, the continental slope, the abyssal plains, and the ocean deeps**.

Continental Margin:

- Continental margins are **transitional zones between continental shores and deep-sea basins**, comprising the **continental shelf, continental slope, continental rise, and deep-oceanic trenches**.
- These features collectively form the boundary between land and ocean, with the deep-oceanic trenches being of particular interest due to their significant role in the distribution and interaction of oceans and continents.



- These trenches, some of the deepest parts of the ocean, provide critical insights into tectonic activity, subduction zones, and the geological processes that shape the Earth's surface.

Continental Shelf:

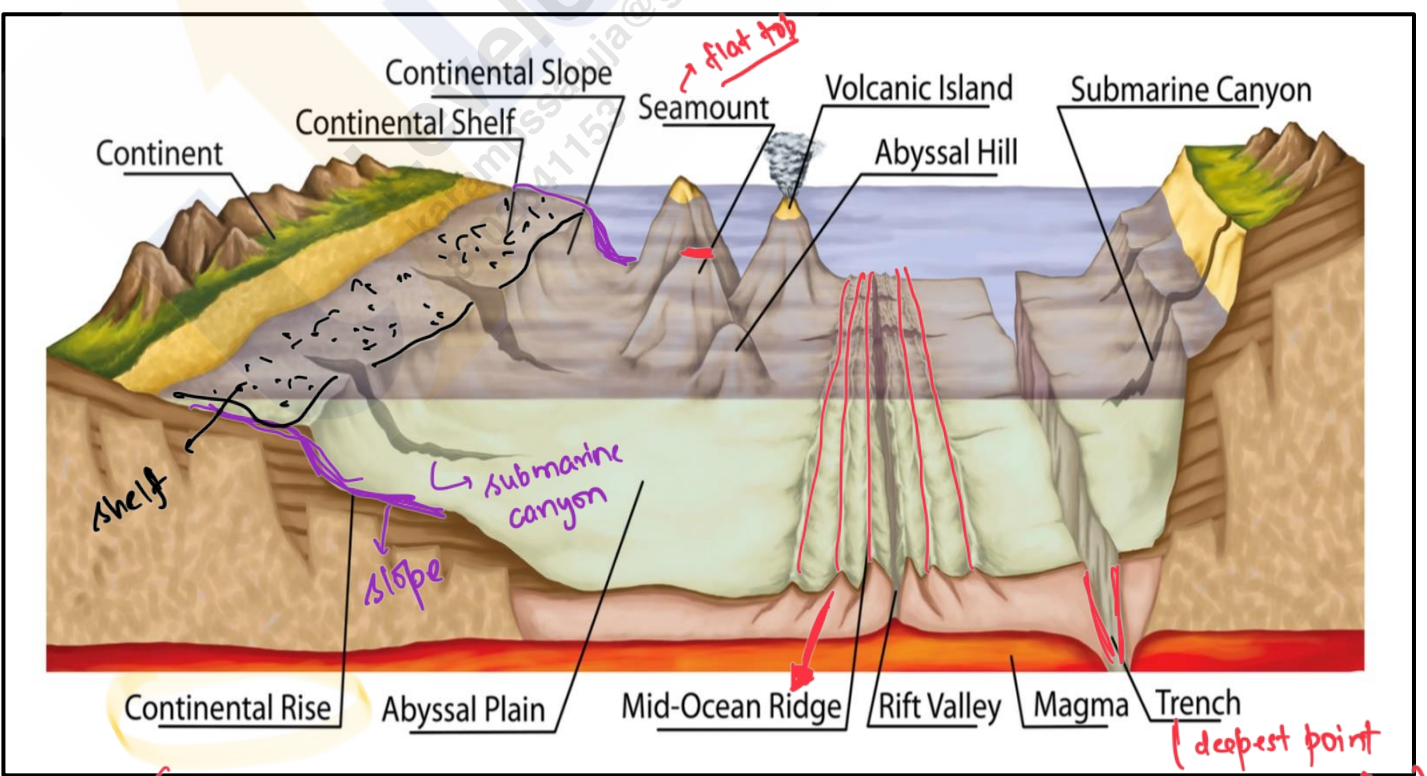
- The continental shelf is a **shallow water area surrounding the main continental landmass**, geologically considered a submerged extension of the continent under oceanic waters.
- Typically, it is **about 100 fathoms deep and gently slopes** towards the ocean at an average rate of 20 meters per Kilometer, with the **angle of slope varying between 1° and 3°**.
- These shelves are narrow submerged landforms **located very close and parallel to the coast**. For example, off the west coast of South America, where the Andes Mountains run close and parallel to the coast, the shelf is only a few hundred meters wide.
- In contrast, the eastern coast of South America features very wide shelves that extend over hundreds of kilometers. On average, a continental shelf is about 70 kilometers wide, but on much of the Pacific Coast, it is relatively narrow, only a few tens of kilometers across.
- The **widest shelves are found in the Arctic Ocean**. A broad continental shelf is the result of sediment build-up shed by a continent.

Shelf Break: → A lot of Resources it have

- The shelf break, also known as the **submerged offshore edge**, marks the point where the shallow continental shelf **transitions to the continental slope**. This boundary is characterized by a noticeable **increase in the steepness** of the sea floor.

Continental Slopes:

- Beginning at the shelf break, the continental slope is characterized by a **steeper angle of slope, ranging from 5° to 60°**. Although it covers only 8.5 percent of the total area of ocean basins, the proportion varies by ocean. It occupies 12.4 percent of the Atlantic Ocean basin, 7 percent of the Pacific Ocean basin, and 6.5 percent of the Indian Ocean basin. Generally, continental slopes are **covered with sediments and gravels**, and they are **not significant sources of mineral deposits**. Features such as **submarine canyons** are commonly associated with continental slopes.



Submarine Canyons:

from cont. shelf, acco. on base of cont. slope.

- Submarine canyons are **long, narrow, and deep valleys or trenches** located on continental shelves and slopes with **steep vertical walls**. These physical features, which start from the continental shelf break and extend to the continental slopes, resemble continental canyons but are situated underwater. Unlike deep-sea trenches found mostly at plate boundaries, undersea **canyons are present along the slopes** of most continental margins.

Continental Rise:

- The continental rise is a **major depositional system in oceans**, consisting of thick sequences of continental material that accumulate between the continental slope and the abyssal plain. These **thick sediments result from three sedimentary processes** - mass wasting, deposition from contour currents, and the **vertical settling** of clastic and biogenic particles.

Abyssal Plains or Deep-Sea Plains:

- Abyssal plains are **extensive, smooth, and flat areas** that lie **between mid-oceanic ridges and continental margins**. These plains are where continental sediments that move beyond the margins are deposited. They are covered with **sediments such as silt and mud**. Abyssal plains are commonly found in the Atlantic Ocean and are unusual in the Pacific Ocean.

Ocean Deepes or Trenches:

- Ocean deepes, also known as trenches, are **depressions on the ocean floors** and represent the **deepest zones of ocean basins**. They are **generally located parallel to coasts with nearby mountains** or along islands, typically in **subduction zones** where oceanic plates converge.

Mid-Ocean Ridges:

- Mid-ocean ridges form along **divergent plate boundaries**, where new ocean floor is created as Earth's tectonic plates spread apart. **Molten rock rises** to the seafloor due to the movement of plates separating from each other. The spreading rate **influences the formation of ridges** - **slower spreading rates result in steeper and more irregular topography**, while **faster rates produce wider profiles with gentler slopes**. Notable examples of mid-ocean ridges include the **Mid-Atlantic Ridge** and the **East Pacific Rise**.

Concept of Sea Floor Spreading:

The concept of sea floor spreading is a pivotal theory in understanding the dynamics of Earth's lithosphere and the processes that drive plate tectonics. Initially proposed by **Harry H. Hess** in the early **1960s**, this theory explains the **creation of new oceanic crust** at mid-ocean ridges and its subsequent movement away from the ridges, leading to the expansion of the ocean floor.

Before the development of the sea floor spreading theory, the **prevailing view was that the ocean floor was static and unchanging**. However, several lines of **evidence began to challenge this view**, including the **discovery of the mid-ocean ridges**, the **symmetrical pattern of magnetic stripes** on either side of these ridges, and the **age distribution of oceanic rocks**.

Older as move away from the crust.

Sea Floor Spreading

- ① Volcanism leads to creation of new ^{ocean} floor. which spreads.
- ②



more spread, more volcanic spread
 more middle ocean ridge are created.



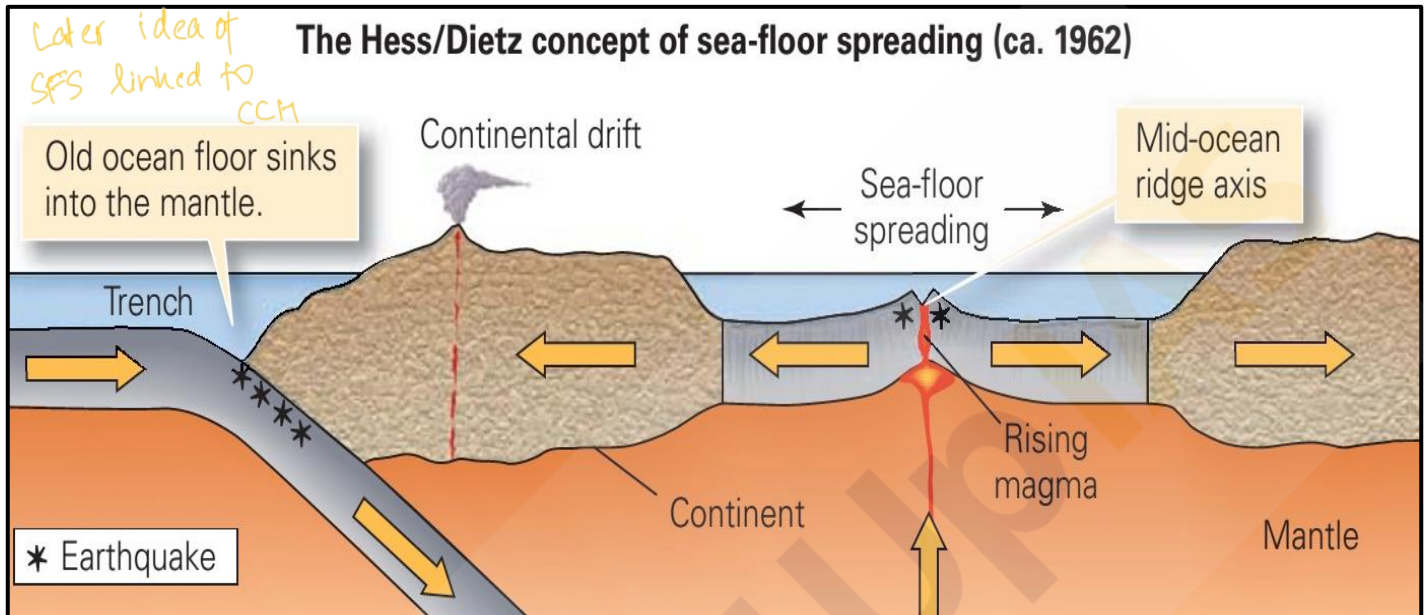
spreading of ocean floor does not lead to exp. of ocean floor
 const. sea floor ocean.

i.e. Ocean floor also getting consumed.



These discoveries led **Harry Hess** to propose the hypothesis known as "**sea floor spreading**" in 1961. Hess argued that **constant eruptions at the crest of oceanic ridges cause the rupture of the oceanic crust, and the new lava wedges into it, pushing the oceanic crust on either side. As a result, the ocean floor spreads.**

The younger age of the oceanic crust, combined with the fact that the spreading of one ocean does not cause the shrinking of another, led Hess to consider the consumption of the oceanic crust. He proposed that the **ocean floor pushed due to volcanic eruptions at the crest sinks down at the oceanic trenches and gets consumed.** This process is a fundamental aspect of the sea floor spreading concept.



Trench → Sinking slab

Mechanism of Sea Floor Spreading:

- **Mid-Ocean Ridges** - The process begins at mid-ocean ridges, which are **underwater mountain ranges formed by upwelling magma from the mantle**. These ridges are characterized by high levels of volcanic activity and are **sites where new oceanic crust is continuously being created**.
- **Upwelling of Magma** - At the mid-ocean ridges, the **upwelling magma rises due to convection currents in the mantle**. As the magma reaches the surface, it cools and solidifies to form new oceanic crust.
- **Movement of Plates** - As **new crust forms, it pushes the older crust away from the ridge, causing the sea floor to spread**. This movement is facilitated by the **divergent boundaries** between tectonic plates.
- **Formation of Ocean Basins** - Over time, the continuous creation and outward movement of oceanic crust led to the **expansion of ocean basins**. This process is responsible for the widening of oceans, such as the Atlantic Ocean.

Evidence Supporting Seafloor Spreading:

- **Magnetic Stripes** - One of the most compelling pieces of evidence for sea floor spreading is the **pattern of magnetic stripes** on the ocean floor. These stripes are **created by the periodic reversal of Earth's magnetic field**, which is recorded in the basaltic rocks as they cool and solidify. The **symmetrical pattern of these stripes on either side of mid-ocean ridges** indicates the continuous creation of new crust and its outward movement.
- **Age of Oceanic Rocks** - The age of oceanic rocks provides further evidence for sea floor spreading. **Rocks near the mid-ocean ridges are relatively young, while those farther away are progressively older**. This age distribution is consistent with the creation of new crust at the ridges and its subsequent movement away.

- Also, the **oceanic crust rocks are significantly younger than the continental rocks**. The age of rocks in the oceanic crust is nowhere more than 200 million years old, whereas some continental rock formations are as old as 3,200 million years. → older oceanic rocks have been destroyed.
- **Heat Flow** - Higher heat flow measurements near mid-ocean ridges compared to the surrounding ocean floor also support the theory of sea floor spreading. The elevated heat flow is attributed to the **upwelling of magma** and the formation of new crust.
- **Distribution of Earthquakes and Volcanoes** - The distribution of **earthquakes and volcanic activity along mid-ocean ridges** aligns with the process of sea floor spreading. These geophysical events are concentrated at the ridges, where the creation and movement of new crust occur.
- **Sediment Thickness on the Ocean Floor** - Contrary to expectations, **the sediments on the ocean floor are very thin**. Scientists anticipated that if the ocean floors were as old as the continents, there would be a complete sequence of sediments for a much longer duration. However, sediment columns older than 200 million years were not found anywhere.

Implications of Sea Floor Spreading:

- **Plate Tectonics** - Sea floor spreading is a **fundamental component of the theory of plate tectonics**, which describes the movement of Earth's lithospheric plates. It provides a mechanism for the movement of oceanic plates and the dynamic nature of Earth's surface.
- **Continental Drift** - The concept of sea floor spreading also **supports the theory of continental drift**, proposed by Alfred Wegener. It explains how continents can move apart as new oceanic crust is created and spreads.
- **Oceanic Features** - Sea floor spreading contributes to the **formation of various oceanic features**, such as **mid-ocean ridges, deep-sea trenches, and transform faults**. It plays a crucial role in the geological and topographical evolution of ocean basins.
- **Global Geological Processes** - Understanding Sea floor spreading helps in comprehending global geological processes, including the **recycling of oceanic crust through subduction**, the **formation of mountain ranges**, and the dynamics of Earth's interior.

Calculation of Sea-Floor Spreading Rates:

The age of magnetic stripes, the rate of sea-floor spreading, and the timing of continent drifting are calculated based on these observations. Magnetic stripes dating back 4.5 million years have been mapped from paleomagnetic surveys of different ocean floors. The rate of sea-floor spreading is calculated using the age of **'isochrons'** (lines joining points of equal dates of magnetic stripes) and the distance between isochrons.

Different oceans have varying spreading rates:

- **The Atlantic and Indian Oceans** spread slowly, at **1.0 to 1.5 cm per year**.
- **The Pacific Ocean** spreads at a faster rate of **6.0 cm per year**.

===== XXXXXX =====



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GEOGRAPHY | - 15

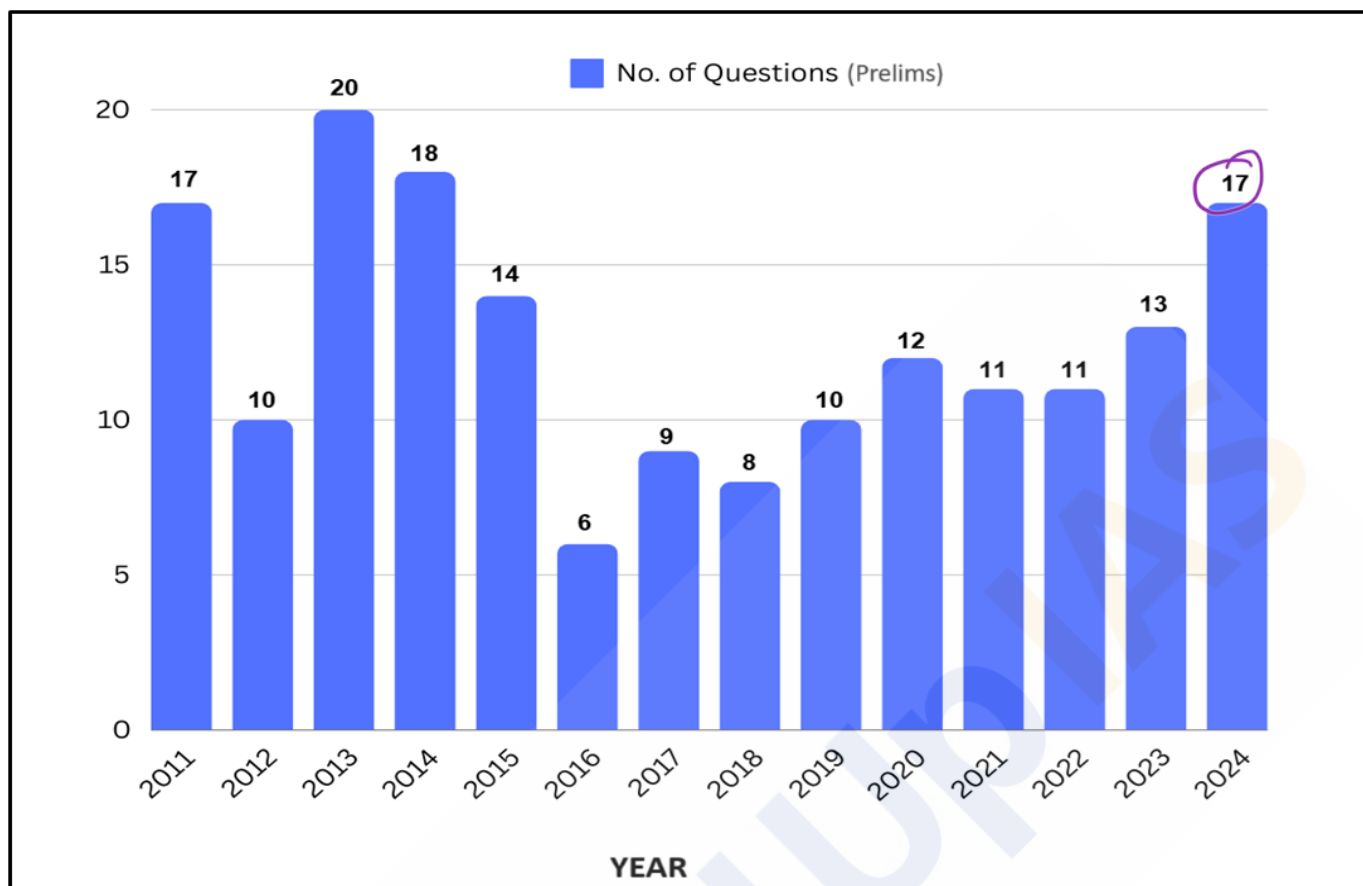
Distribution of Oceans and Continents-II

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Year-wise Trend Analysis (2011 – 2024)



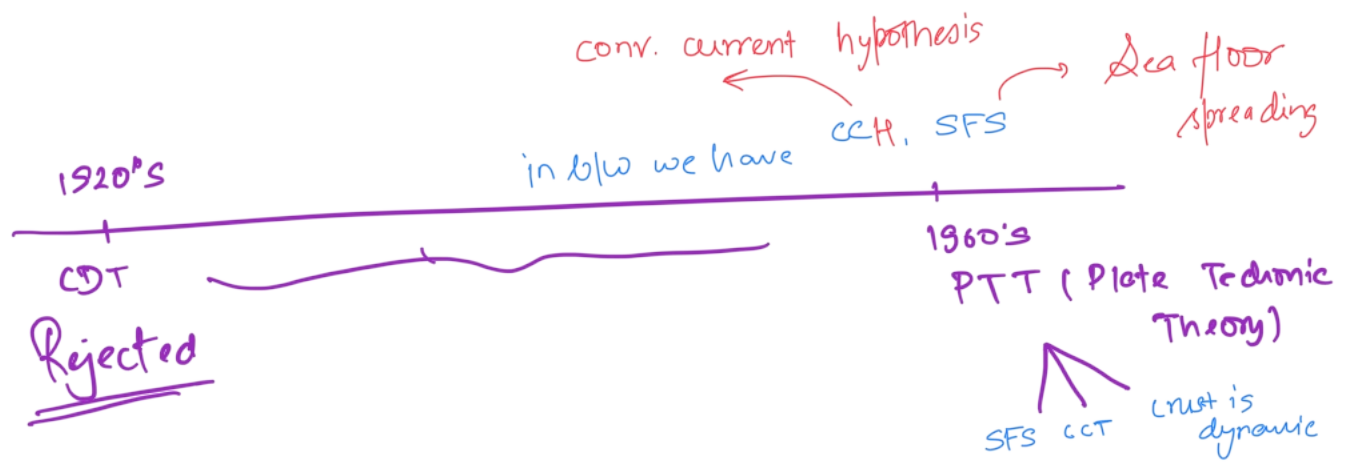
Topic-wise Trend Analysis (2011 – 2024)

Physical Geography

Topic	Questions
Geomorphology	9
Oceanography	5
Climatology	22
Mapping	16
Distribution of Resources	5
Miscellaneous	15

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Topic	Questions
Physiography of India	24
Drainage System of India	15
Indian Climate	7
Indian Soils	5
Natural Vegetation of India	12
Indian Agriculture	31
Mineral Resources of India	10



conv. current (4 parts)

- ↳ Rising limb
- ↳ Diverging limb
- ↳ Sinking limb
- ↳ converging limb

- ① convergent boundary
→ Pacific circumference
- ② Divergence → MAO
→ Antarctic
→ MIO
→ East Pacific
- ③ Transform boundary → California
→ Turkey

Plate Tectonics Theory

Introduction:

The term "**plate tectonics**" was first coined by **Tuzo Wilson** of the **University of Toronto**. This scientific theory describes the **large-scale motion of seven large plates and numerous smaller plates** in the Earth's lithosphere over hundreds of millions of years.

The theoretical model of plate tectonics **builds on the concept of continental drift**, which was developed during the first few decades of the 20th century. The geoscientific community widely accepted plate tectonics after the validation of seafloor spreading in the late 1950s and early 1960s. **McKenzie and Parker** suggested the theory in **1967**, with **Morgan** outlining it further in **1968**.

Development of the Theory:

The comprehensive Plate Tectonics theory aims to explain most of the dynamism of the Earth's crust and the features formed by endogenetic forces. The development of this theory began in the **1960s** with extensive seafloor mapping.

The theory is based on two principal hypotheses:

- **Arthur Holmes' convection current hypothesis.**
- **The concept of seafloor spreading advocated by Harry Hess.**

It is an **improvement over Wegener's continental drift theory** and is considered the most sophisticated and comprehensive explanation for the drift of continents and the expansion of seafloors.

According to this Theory:

According to the Plate Tectonics theory, **the lithosphere**, which includes the **crust and the uppermost part of the mantle**, is broken into fragments known as tectonic plates. These **plates float on a ductile layer** called the **asthenosphere**, which is located in the uppermost part of the mantle. The lithosphere's thickness varies, ranging from 5-100 km in oceanic regions to about 200 km in continental areas.

The movement of these plates is attributed to **convection currents** generated in the upper mantle. These currents cause the **plates to move horizontally over the asthenosphere as rigid units**. The lithospheric plates can be categorized as **oceanic plates**, which are thinner, and **continental plates**, which are thicker.

The movement of these crustal plates due to convection currents in the mantle causes **various landforms to form and is the principal cause of all earth movements**. The margins of the plates are sites of **significant geologic activity**, including seafloor spreading, volcanic eruptions, crustal deformation, mountain building, and continental drift.

Lithospheric Plates:

A lithospheric plate (tectonic plate) is a broad segment of the lithosphere that floats on the underlying asthenosphere and moves independently of other plates. These plates can be broadly classified into continental plates and oceanic plates. The theory of plate tectonics proposes that the earth's lithosphere is divided into **seven major and some minor plates**.

Major Plates:

- Antarctica and the surrounding oceanic plate
- North American plate
- South American plate
- Pacific plate
- India-Australia-New Zealand plate
- Africa with the eastern Atlantic floor plate
- Eurasia and the adjacent oceanic plate

Minor Plates:

- Cocos plate (Between Central America and Pacific plate)
- Nazca plate (Between South America and Pacific plate)
- Arabian plate (Mostly the Saudi Arabian landmass)
- Philippine plate (Between the Asiatic and Pacific plate)
- Caroline plate (Between the Philippine and Indian plate (North of New Guinea))
- Fuji plate (North-east of Australia)
- Turkish plate
- Aegean plate (Mediterranean region)
- Caribbean plate
- Juan de Fuca plate (between Pacific and North American plates)
- Iranian plate.

There are **many more minor plates** other than the ones mentioned above. Most of these minor plates were formed due to stress created by converging major plates.



Interaction of Tectonic Plates:

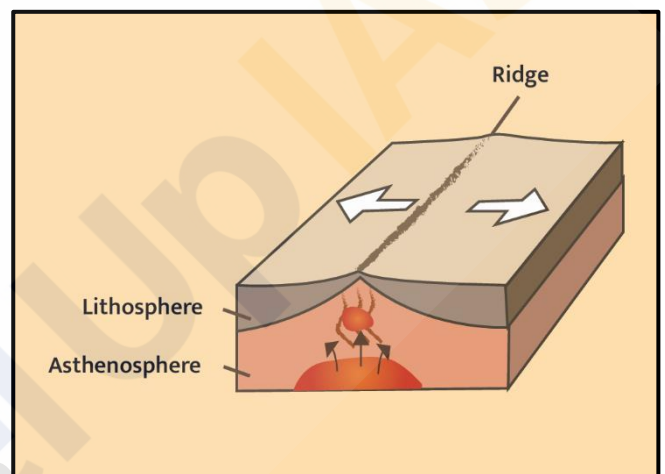
The Earth's surface is shaped by the interaction of various tectonic plates, leading to the formation of major geomorphological features such as fold and block mountains, mid-oceanic ridges, trenches, volcanism, and earthquakes.

There are three ways in which the plates interact with each other:

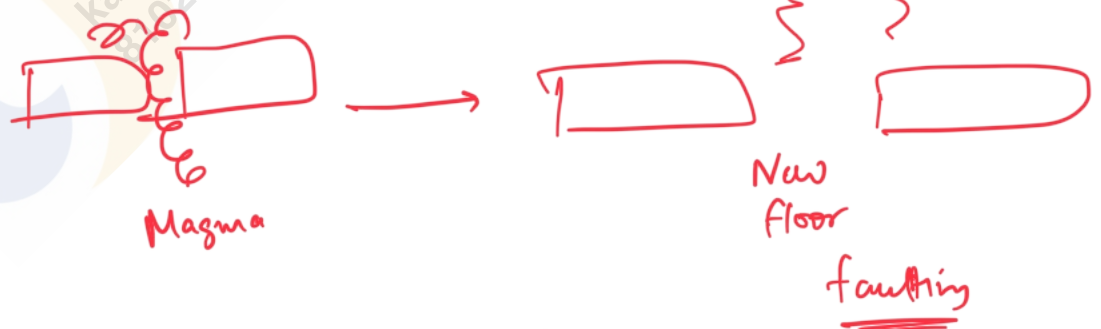
- **Divergent** or **Constructive** Plate Boundaries.
- **Convergent** or **Destructive** Plate Boundaries.
- **Transform** or **Conservative** Plate Boundaries.

Divergent or Constructive Plate Boundaries:

- Divergent boundaries occur **when two tectonic plates move away from each other**. At these boundaries, **magma from the mantle rises through long fissures, creating new oceanic crust** as it solidifies into basalt.
- This process is accompanied by **frequent earthquakes** and the formation of **mid-ocean ridges** (e.g. Mid-Atlantic Ridge), **rift valleys** (e.g. East African Rift), and **fissure volcanoes**.
- Divergent boundaries are sites of **earth crust formation** (hence the name constructive edge), and volcanic earth forms are common along such edges. **Earthquakes** (shallow focus) are also common along divergent edges.



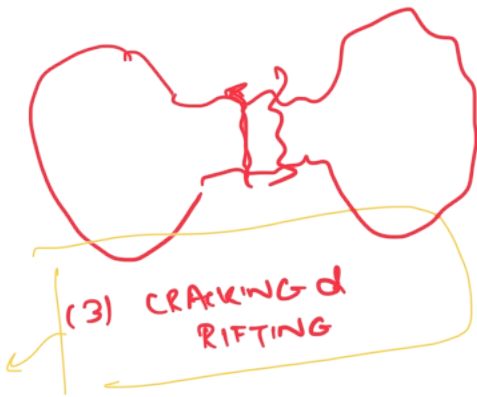
- Lf →
- (1) Volcanic
 - (2) Earthquake
 - (3) Mid ocean ridge
 - (4) ~~MOR~~ New Ocean floor
 - (5) Normal fault, Rifting
 - (6) Transform, fault



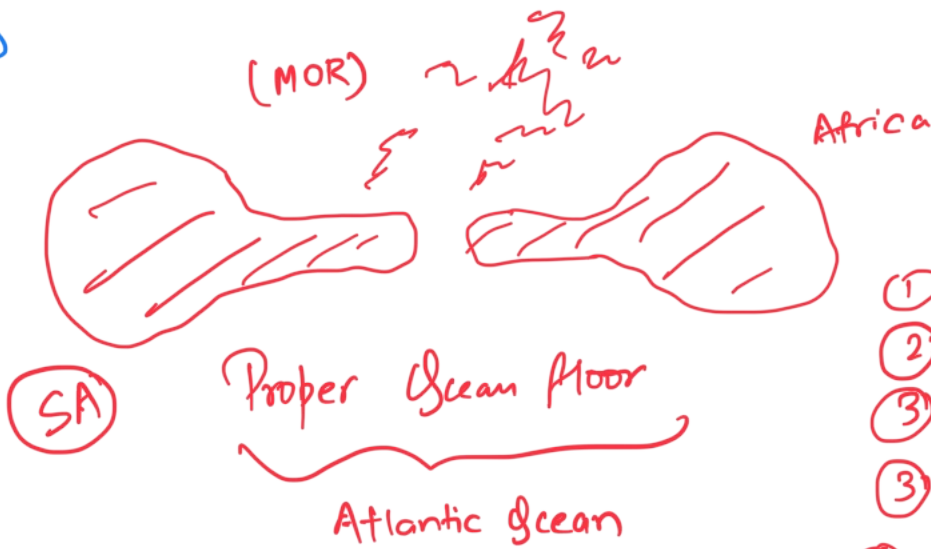
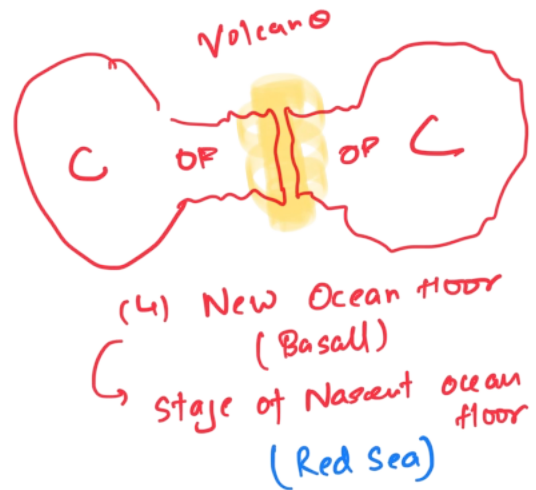
Transverse fault → slipping plate Boundary

Reverse fault → convergence Plate Boundary

Divergent Plate Boundary → below continent

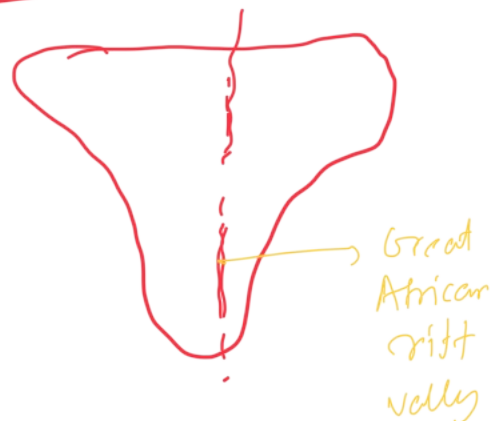


Great African rift valley



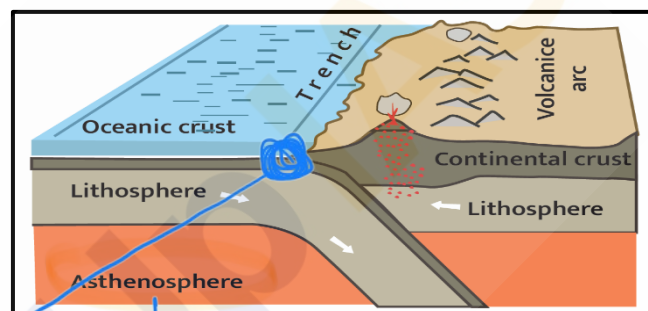
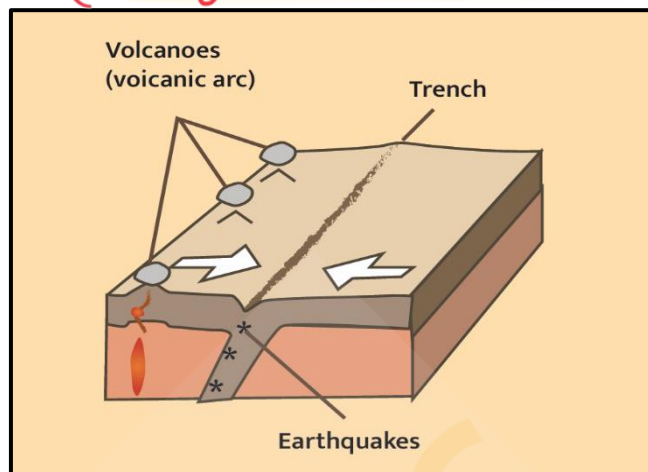
- ① EQ
- ② Volcano
- ③ Rift valley
- ③ Fault
- ④ Transform fault
- ⑤ MOR

Seen near Great Africa



Convergent or Destructive Plate Boundaries: (convergent limb & sinking limb)

- Convergent boundaries are formed **when two tectonic plates collide**. The resulting impact can lead to the creation of **rugged mountain ranges** and **deep-sea trenches**.
- Volcanic activity** often occurs along these boundaries, with magma rising from the melting plates to form new crust.
- Near the convergent edge a part of the **crust is destroyed**, hence the name **Destructive Edge**. The subducted material gets heated, up and is thrown out forming **volcanic island arc** and **continental arc systems**.
- There are three types of Convergent Boundaries:**



1. Subduct OF → Trench

2. Vol. (3) EG

3. Fold Mountain

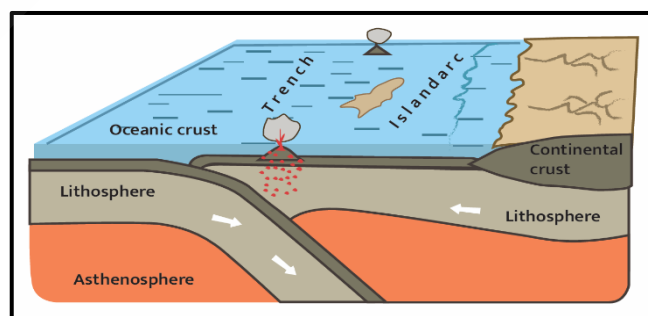
4. Volcanic Peak

Accretionary Wedge

→ Fold Mountains { Most scrapping b/w ocean & continent }

In Asthenosphere crust melted volcanism happens.

- Oceanic-Oceanic (O-O) Convergence** - When two oceanic plates collide, the **denser plate subducts**, forming an **oceanic trench** and a **volcanic island arc**. Over time, this can lead to the development of a mature island arc system, such as Japan or the islands of Sumatra and Java in Indonesia.

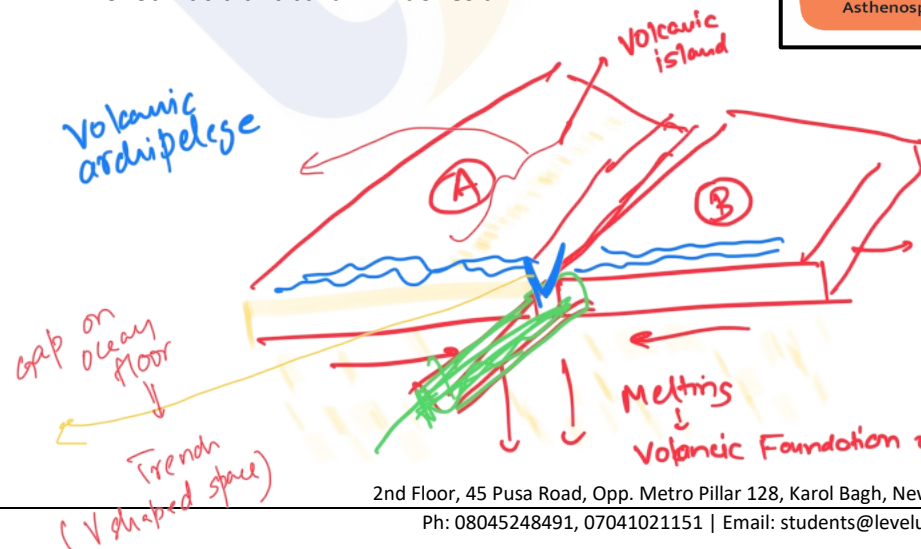


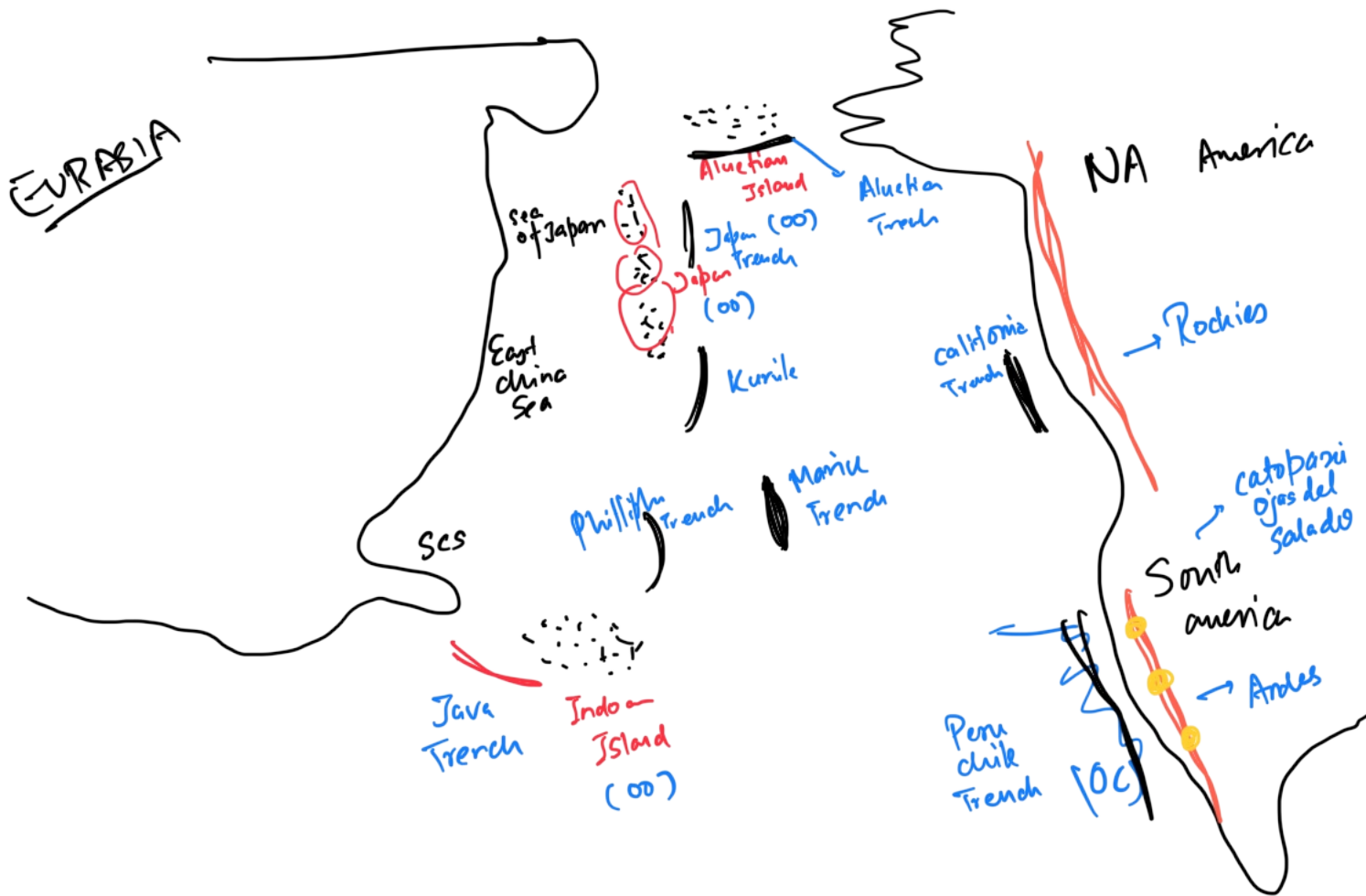
(1) Trench
(2) EG

(3) Volcanism

(4) Vol. island (archipelago)

B ocean floor subducted in mantle due to conv. current



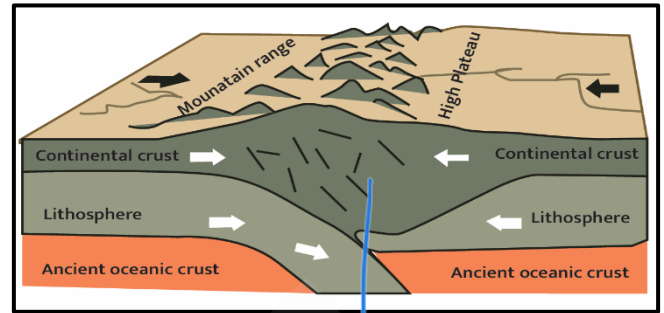


→ Sediment of Ocean floor on edge of other ocean floor forming Accretionary Ridge.

→ Himalaya is not a volcanic mountain but still has some volcanic rock, due to older sediment of oo & oc collision.

→ Alps due to Eurasia & African Tectonic plate in Europe.

- **Continental-Continental (C-C) Convergence** - This type of convergence occurs between two continental plates, leading to the **upthrust of oceanic sediments** and the **formation of fold mountains**, exemplified by the Himalayan Mountains. **Both the plates have a density that is much lower than the mantle, which prevents subduction.**

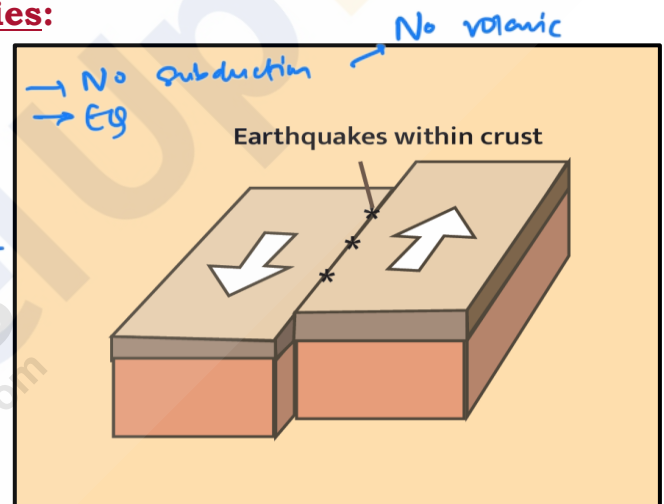


- es Himalayas, Hindukush, Urals, Alps } CC collision mountain
- (1) No subduction
(2) No trenches
(3) No volcanic
(4) EQ.
(5) Doubling of crust (6) Fold Mountains
- Accretion wedge
↓
Fold Mountain

★ Plate Tectonic visualize convergence of boundary in a sequence like OC convergence followed by OC conv. followed by CC conv.

Transform or Conservative Plate Boundaries:

- A transform fault or transform boundary, sometimes called a '**strike-slip boundary**,' is a fault along a plate boundary where the **motion is predominantly horizontal**.
- Transform plate boundaries are locations where **two plates slide past each other without the creation or destruction of landforms but only deformation of existing landforms**. The fracture zone that forms a transform plate boundary is known as a **transform fault**.
- In oceans, transform faults are the planes of separation generally perpendicular to the mid-oceanic ridges.
- **San Andreas Fault** (Silicon Valley lies dangerously close to the Faultline) along the western coast of USA is the best example for a transform boundary on continents.



eg. Juan de Fuca Plate collide with NA American Plate.

Evidence in Support of the Plate Tectonic Theory:

The theory of plate tectonics, which describes the large-scale movement of Earth's lithosphere, is supported by a multitude of **evidence derived from various geological, geophysical, and oceanographic studies**. Below are the detailed explanations of key evidences that bolster the plate tectonic theory.

Ocean Deep Drilling - The Glomar Challenger:

- One of the most compelling pieces of evidence for plate tectonics comes from ocean deep drilling projects, notably the **Glomar Challenger expedition**. Launched in **1968**, the Glomar Challenger was a **deep-sea research and scientific drilling vessel**. It conducted **extensive drilling into the ocean floor**, collecting core samples from various locations across the world's oceans. **Analysis of these cores revealed significant findings:**
 - **Age of Oceanic Crust** - The core samples showed that the **age of the oceanic crust increases with distance from mid-ocean ridges**, consistent with the idea that new crust is formed at these ridges and then moves outward.

- **Sediment Thickness** - The **thickness of sediments on the ocean floor increases with distance from the ridges**, supporting the notion of seafloor spreading where new crust forms and older crust moves away, accumulating more sediment over time.
- **Magnetic Anomalies** - The cores provided data on the magnetic properties of the ocean floor, further corroborating the patterns of **magnetic reversals** recorded in the crust.

JOIDES - Joint Oceanographic Institutions for Deep Earth Sampling:

- The JOIDES program, which stands for Joint Oceanographic Institutions for Deep Earth Sampling, continued the work begun by the Glomar Challenger. This international collaboration aimed to explore the structure and history of Earth's crust beneath the ocean. **Key contributions from JOIDES include:**
 - **Subsurface Geology** - Drilling operations revealed details about the subsurface geology, including the **composition and structure of oceanic crust** and the underlying mantle.
 - **Plate Boundaries** - The program provided crucial data on the nature of plate boundaries, such as **mid-ocean ridges, subduction zones, and transform faults**.
 - **Hydrothermal Vents** - Discoveries of **hydrothermal vent systems along mid-ocean ridges** offered insights into the thermal and **chemical exchanges** between the oceanic crust and seawater.

Hot Spots: (within Plate volcanic Region, They are fixed where magma comes from deep interior. These hot spot are known to -

- Hot spots are another critical piece of evidence for plate tectonics. These are **volcanic regions thought to be fed by underlying mantle** that is anomalously hot compared to the surrounding mantle. **Key points include:**
 - **Fixed Mantle Plumes** - Unlike the movement of tectonic plates, **hot spots are generally stationary** and provide a fixed reference point. The **Hawaiian Islands** are a prime example, where a chain of volcanic islands formed as the Pacific Plate moved over a hot spot. → Mid Way → Emperor Sea Mount
 - **Age Progression** - The **age of the islands increases progressively from southeast to northwest**, reflecting the **plate's movement direction**. This pattern provides strong evidence for plate motion.

Paleomagnetism:

- Paleomagnetism, the study of the record of Earth's magnetic field in rocks, sediments, and archaeological materials, provides strong evidence for plate tectonics. **Key aspects include:**
 - **Magnetic Minerals** - As volcanic rocks cool, **magnetic minerals within them align with Earth's magnetic field**, locking in a record of the **field's direction and intensity** at that time.
 - **Apparent Polar Wander** - The study of paleomagnetic data from different continents showed that the **magnetic poles appeared to wander over time**. This apparent polar wander can be explained by the movement of continents rather than movement of the poles themselves.
 - **Magnetic Reversals** - The record of **magnetic reversals in oceanic crust** matches across different ocean basins, indicating symmetric spreading from mid-ocean ridges.

Magnetic Reversal and Seafloor Spreading:

- The phenomenon of magnetic reversal and seafloor spreading is crucial to understanding plate tectonics. **Important points include:**
 - **Magnetic Stripes** - The ocean floor displays **symmetrical stripes of normal and reversed magnetic polarity on either side of mid-ocean ridges**, created as new crust forms and records the Earth's magnetic field.

- **Age Dating** - Dating of these magnetic stripes shows that they correspond to known periods of magnetic reversal, providing a **timeline for seafloor spreading**.
- **Spreading Rates** - The **symmetrical nature and consistent age progression** of these magnetic stripes provide estimates of the rates at which different sections of the seafloor are spreading.

Evidence from Continental Drift Theory:

- The **Continental Drift Theory, proposed by Alfred Wegener**, provided initial evidence that later supported plate tectonics. **Relevant points include:**
 - **Fit of the Continents** - The **jigsaw-like fit of continents**, particularly noticeable between South America and Africa, suggests they were once joined and have drifted apart.
 - **Fossil Correlation:** Identical **fossils of plants and animals found on widely separated continents** indicate that these lands were once connected.
 - **Geological Continuity** - **Similar rock formations and mountain ranges on different continents**, such as the Appalachian Mountains in North America and the Caledonian Mountains in Scotland and Scandinavia, suggest these continents were once part of a single landmass.
 - **Glacial Evidence** - Evidence of **past glaciations in currently tropical regions**, such as India and Africa, indicates these continents were once located closer to the poles.

Rates of Plate Movement:

- Scientists determine the rates of plate movement by examining the **patterns of normal and reverse magnetic fields that parallel the mid-oceanic ridges**. These magnetic strips provide crucial evidence about the speed at which tectonic plates are moving. The rates of plate movement vary significantly across different regions.
- For instance, the **Arctic Ridge, located in the Arctic Ocean**, exhibits the **slowest rate of plate movement**, measuring less than **2.5 centimeters per year**.
- In contrast, the **East Pacific Rise** near Easter Island, situated in the South Pacific about 3,400 kilometers west of Chile, shows the **fastest rate of plate movement, exceeding 15 centimeters per year**. This variation in plate movement rates is essential for understanding the dynamic nature of Earth's lithosphere.

Improvement of Plate Tectonics over Continental Drift Theory:

Mechanism of Movement:

- **Continental Drift Theory:** Proposed by Alfred Wegener in 1912, it suggested that continents drift across the Earth's surface. However, it lacked a convincing mechanism explaining how continents could move through solid oceanic crust.
- **Plate Tectonics Theory:** Introduced in the 1960s, it explains that the Earth's lithosphere is divided into several plates that float on the semi-fluid asthenosphere beneath. This theory provides a robust mechanism, driven by **mantle convection, slab pull, and ridge push, explaining the movement of plates**.

Evidence Integration:

- **Continental Drift Theory:** Wegener's evidence included the fit of continental coastlines (e.g., South America and Africa), fossil distribution, and geological similarities across continents. While compelling, it was largely circumstantial.

- **Plate Tectonics Theory:** It integrates a wide range of evidence, including the **distribution of earthquakes** and **volcanoes**, the **age and spreading of ocean floors**, and the **matching geological features across plate boundaries**. The discovery of mid-ocean ridges and deep-sea trenches, along with paleomagnetic studies, provided solid proof.

Global Framework:

- **Continental Drift Theory:** Focused primarily on the movement of continents and did not address the interactions between oceanic and continental crust.
- **Plate Tectonics Theory:** Offers a comprehensive framework for understanding the dynamic nature of the entire lithosphere. It explains the **formation of mountains, ocean basins**, earthquakes, and volcanoes through the interactions of tectonic plates, encompassing both continental and oceanic crust.

Predictive Power:

- **Continental Drift Theory:** Was largely descriptive and did not offer predictive models for future geological events.
- **Plate Tectonics Theory:** Provides predictive models for seismic activity, volcanic eruptions, and the creation of geological features. It allows scientists to understand and **anticipate changes in Earth's surface over geological time scales**.

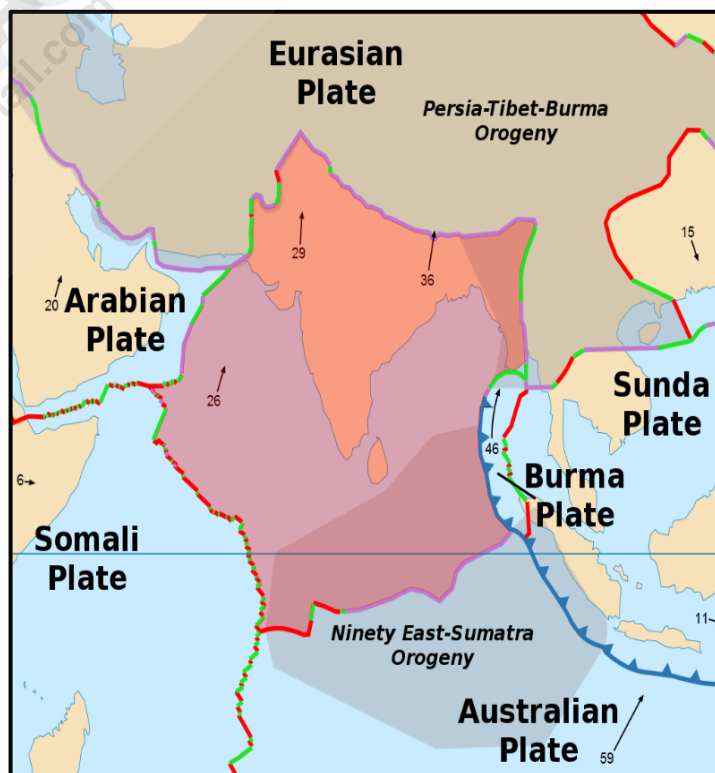
Technological Advancements:

- **Continental Drift Theory:** Developed in the early 20th century, it lacked the technological tools and data for extensive validation.
- **Plate Tectonics Theory:** Benefited from advancements in technology, such as **seismic tomography**, **satellite geodesy**, and **deep-sea exploration**, which provided concrete data supporting the theory.

Movement of the Indian Plate:

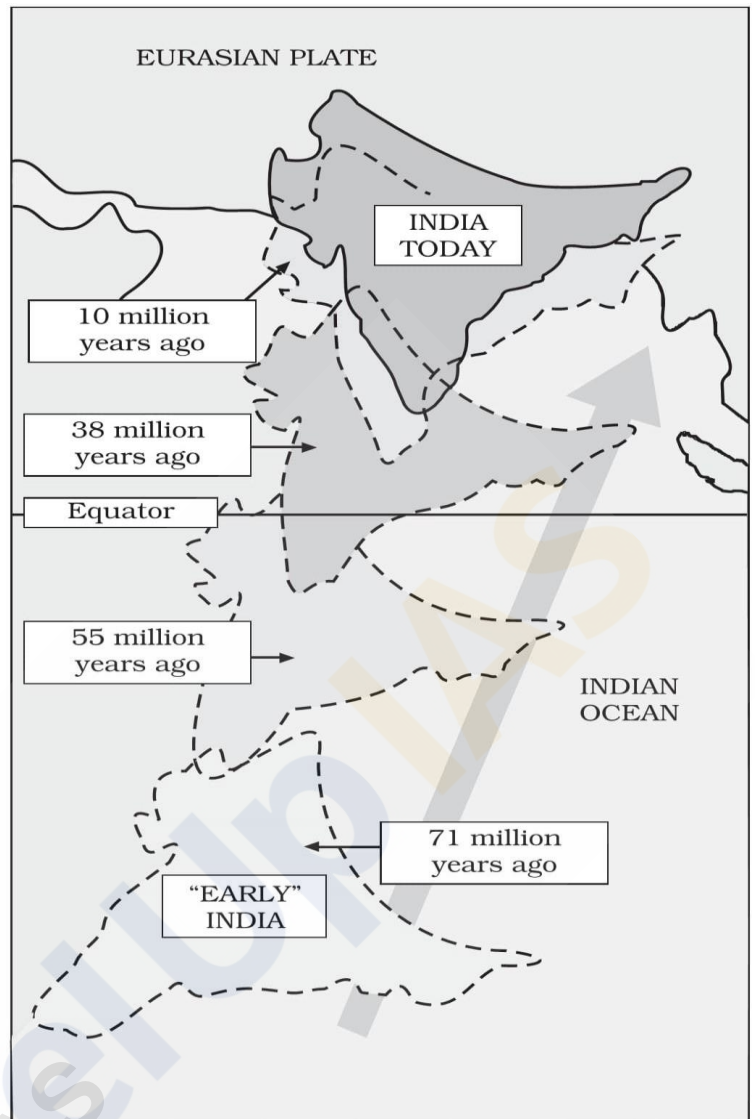
The Indian Plate:

- The Indian Plate comprises **Peninsular India** and **parts of the Australian continent**. The **northern boundary** of this plate is marked by the **subduction zone along the Himalayas**, indicating a continent-continent convergence.
- On the **eastern side**, the plate extends through the **Rakinyoma Mountains of Myanmar** towards the island arc along the **Java Trench**. This eastern margin is a spreading site located east of Australia, represented by an oceanic ridge in the southwestern Pacific.
- The **western boundary** follows the **Kirthar Mountain range in Pakistan**, continuing along the Makrana coast, and connects with the spreading site from the **Red Sea rift**, extending southeast along the **Chagos Archipelago**.
- Additionally, the boundary between the Indian and Antarctic plates is marked by an **oceanic ridge** (a divergent boundary) running approximately in a west-east direction, merging into the spreading site south of New Zealand.



Movement and Collision:

- India was **once a large island positioned off the coast of Australia** in a vast ocean, **separated from the Asian continent by the Tethys Sea around 225 million years ago.**
- The **northward journey** of the Indian Plate began approximately **200 million years ago**, coinciding with the **breakup of Pangaea.**
- The **collision with Asia occurred around 40-50 million years ago**, leading to the rapid uplift of the Himalayas.
- The positions of India since around 71 million years ago to the present reveal significant geological changes. About **140 million years ago, the Indian subcontinent was located at roughly 50°S latitude.**
- The **Tethys Sea separated the two major plates, with the Tibetan block closer to the Asiatic landmass.**
- As the Indian Plate moved northward towards the Eurasian Plate, a significant event was the **outpouring of lava and the formation of the Deccan Traps around 60 million years ago**, continuing for an extended period. During this time, the subcontinent remained close to the equator.
- From around **40 million years ago** and continuing thereafter, the **formation of the Himalayas** took place. Scientists believe that this process is ongoing, with the height of the Himalayas still increasing to this day.



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UPSC Previous Year Questions (Mains)

2021 – Q. Mention the global occurrence of volcanic eruptions in 2021 and their impact on regional environment.

(10 Marks, 150 Words)

2020 – Q. Discuss the geophysical characteristics of the Circum-Pacific Zone.

(10 Marks, 150 Words)

2018 – Q. Define mantle plume and explain its role in plate tectonics.

(10 Marks, 150 Words)

2014 – Q. Why are the world's fold mountain systems located along the margins of continents? Bring out the association between the global distribution of fold mountains and earthquakes and volcanoes.

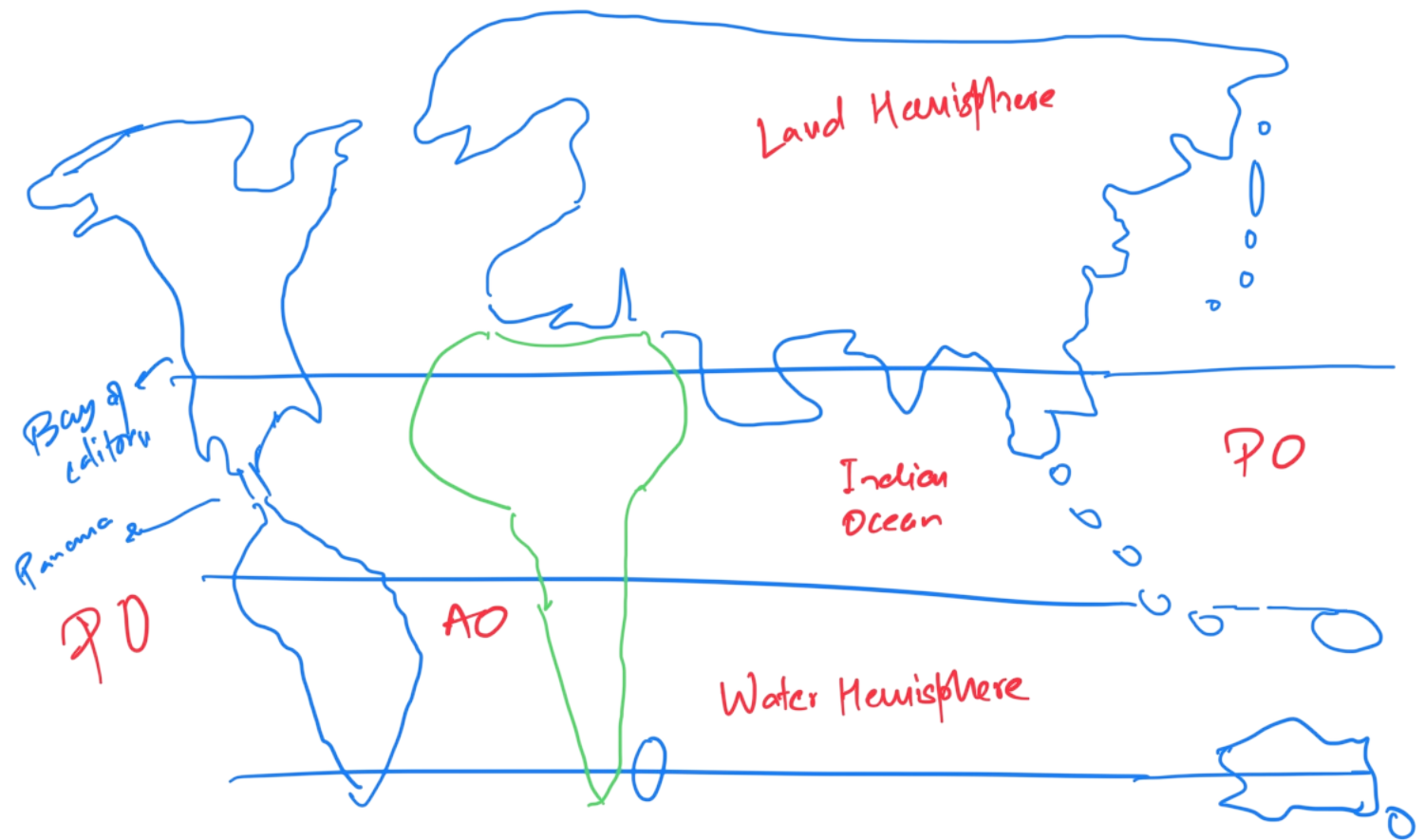
(10 Marks, 150 Words)

2014 – Q. Explain the formation of thousands of islands in the Indonesian and Philippines archipelagos.

(10 Marks, 150 Words)

Draw World Map

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GS FOUNDATION 2024-25

GEOGRAPHY - 16

Minerals and Rocks

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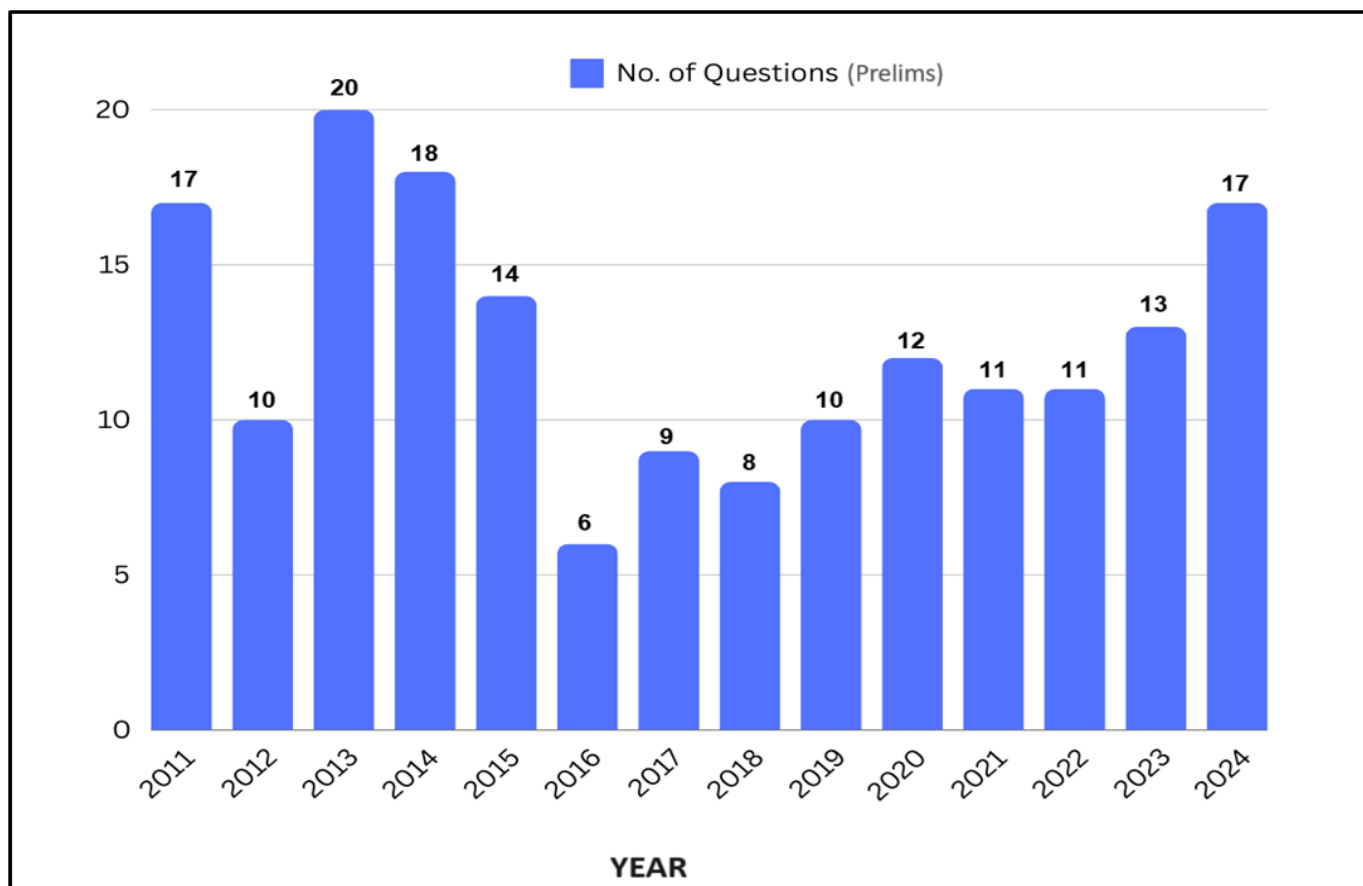
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Minerals

Introduction:

The Earth is composed of various elements, existing in solid form on the outer layer and in hot, molten form in the interior. **About 98% of the Earth's crust is made up of eight primary elements - oxygen, silicon, aluminium, iron, calcium, sodium, potassium, and magnesium.** The remaining 2% consists of elements such as titanium, hydrogen, phosphorus, manganese, sulphur, carbon, nickel, and others.

The elements in the earth's crust are rarely found exclusively but are usually **combined with other elements to make various substances.** Though the lithosphere contains a limited number of elements, these elements combine in numerous ways, resulting in a wide variety of minerals. **Over 2,000 minerals have been identified in the Earth's crust,** but the **majority belong to six major rock-forming mineral groups.**

The primary source of all minerals is the hot magma within the Earth. As magma cools, mineral crystals form in a specific sequence, solidifying to create rocks. Minerals such as coal, petroleum, and natural gas are organic and found in solid, liquid, and gaseous forms, respectively.

Mineral

A mineral is a naturally occurring substance that can be either organic or inorganic. It possesses an orderly atomic structure, along with a specific chemical composition and distinct physical properties. Typically, a mineral consists of two or more elements. However, there are instances where minerals consist of single elements, such as sulphur, copper, silver, gold, and graphite.

Physical Characteristics of Minerals:

External Crystal Form:

- The external crystal form of a mineral is **determined by its internal molecular arrangement.** This can result in various geometric shapes such as **cubes, octahedrons, and hexagonal prisms.**

Cleavage:

- Cleavage refers to **the tendency of minerals to break along specific planes,** resulting in relatively flat surfaces. This is also a consequence of the **internal arrangement of molecules.** Minerals may cleave in one or more directions and at any angle relative to each other.

Fracture:

- When the internal **molecular arrangement of a mineral is too complex** to form planes of cleavage, it **breaks in an irregular manner.** This type of breakage is known as fracture.

Lustre:

- Lustre describes the **appearance of a mineral's surface without considering its colour.** Each mineral has a distinctive lustre, such as metallic, silky, or glossy.

Colour:

- The colour of a mineral can be a characteristic feature determined by its molecular structure. For instance, malachite, azurite, and chalcopyrite have characteristic colours. Additionally, some minerals may be coloured by **impurities,** as seen with quartz, which can appear white, green, red, or yellow depending on the impurities present.

Streak:

- Streak refers to the **colour of the powdered form of a mineral**. This may be the same as the mineral's apparent colour or different. For example, malachite has a green streak, while fluorite, despite being purple or green, gives a white streak.

Transparency

- Transparency is classified into **three categories** - **transparent**, where light passes through the mineral allowing objects to be seen clearly; **translucent**, where light passes but is diffused, making objects appear blurred; and **opaque**, where no light passes through.

Structure:

- The structure of a mineral pertains to the **specific arrangement of its individual crystals**. This can be fine, medium, or coarse-grained. Additionally, some minerals may exhibit fibrous structures that can be separable, divergent, or radiating.

Hardness:

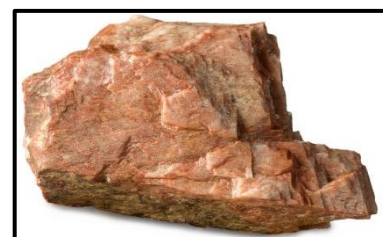
- Hardness measures a **mineral's resistance to being scratched**. This property is quantified on a **scale from 1 to 10** using a reference list of minerals: 1. Talc, 2. Gypsum, 3. Calcite, 4. Fluorite, 5. Apatite, 6. Feldspar, 7. Quartz, 8. Topaz, 9. Corundum, and 10. Diamond. For context, a fingernail has a hardness of 2.5, while glass or a knife blade is about 5.5.

Specific Gravity

- Specific gravity is the **ratio of the weight of a mineral to the weight of an equal volume of water**. This property is determined by weighing the mineral in air and then in water, and dividing the weight in air by the difference between the two weights.

Major Minerals and Their Characteristics:**Feldspar:**

- Feldspar is **the most abundant mineral on Earth**, constituting nearly **half of the Earth's crust**. It is composed primarily of **silicon, oxygen**, and varying amounts of **sodium, potassium, calcium, and aluminium**. Feldspar exhibits a **light cream to salmon pink colour** and is widely used in ceramics and glass manufacturing.

**Quartz:**

- Quartz is another essential mineral found in **abundance in rocks like sand and granite**. It is composed of **silica** and is characterized by its hardness and **insolubility in water**. Quartz is typically **colourless or white** and has diverse applications, including in the radio and radar industries.

**Pyroxene:**

- Pyroxene is a common mineral containing **calcium, aluminium, magnesium, iron, and silica**. It accounts for approximately **10% of the Earth's crust** and is even found in **meteorites**. Pyroxene minerals often exhibit **green or black coloration**.



Amphibole:

- Amphiboles are a group of minerals containing **aluminium, calcium, silica, iron,** and **magnesium**. They constitute about **7% of the Earth's crust** and are commonly found in **green or black varieties**. **Asbestos**, a commercially important material, is derived from amphiboles. **Hornblende** is a well-known example of an amphibole mineral.

**Mica:**

- Mica is a mineral composed of **potassium, aluminium, magnesium, iron,** and **silica**. It forms approximately **4% of the Earth's crust** and is commonly found in igneous and metamorphic rocks. Mica's unique properties make it valuable in the electrical industry.

**Olivine:**

- Olivine is a mineral rich in **magnesium, iron,** and **silica**. Often found in **greenish crystals**, it is commonly associated with basaltic rocks. While not as abundant as some other minerals, olivine has applications in the **jewellery industry**.

**Other Important Minerals:**

- Beyond these primary minerals, several other minerals like **chlorite, calcite, magnetite, hematite, bauxite,** and **barite** are present in varying quantities within rocks. These minerals contribute to the overall composition and properties of the rock formations.

Classification of Minerals:

Minerals can be broadly classified into **two categories - metallic and non-metallic minerals**. This classification is based on the presence or absence of metal content in the minerals.

Metallic Minerals:

Metallic minerals **contain metal content** and are further **subdivided into three types** based on their specific properties and uses:

- Precious Metals** - These are rare and **valuable metals**, including **gold, silver,** and **platinum**. They are often used in jewellery, electronics, and as investment assets.
- Ferrous Metals** - This group includes **iron and other metals that are typically mixed with iron** to create various kinds of steel. These metals are crucial in the **construction and manufacturing industries** due to their strength and durability.
- Non-Ferrous Metals** - These metals **do not contain iron** such as **copper, lead, zinc, tin,** and **aluminium**. They are used in a wide range of applications, from electrical wiring (copper) to packaging (aluminium) and batteries (lead).

Non-Metallic Minerals:

Non-metallic minerals **do not contain metal content** and have a diverse range of applications. Examples include:

- Sulphur** - Used in the production of **sulfuric acid**, which is a key industrial chemical.
- Phosphates** - Essential for the manufacture of fertilizers, which are critical for agricultural productivity.
- Nitrates** - Also used in **fertilizers** as well as in **explosives**.

Additionally, **cement is a notable product made from a mixture of non-metallic minerals**. It is a fundamental material in the construction industry, used for making concrete and other building structures.

Important Metallic Minerals:

Iron Ore:

Iron ore is a term used to describe rocks and minerals from which metallic iron can be economically extracted. These **ores are rich in iron oxides** and are typically found in various colours such as **dark grey, bright yellow, rusty red, and deep purple**. Iron ores are classified based on the percentage of metallic iron present:

- **Magnetite** - The best quality iron ore, containing more than 70% metallic iron. It is black and exhibits magnetic properties, making it highly valuable.
- **Hematite** - This ore contains about 65-70% metallic iron and is usually red or brown in colour. It is one of the primary ores used for iron extraction.
- **Siderite** - Siderite has a lower iron content, about 40-50%, and contains more impurities. It is brown and consists of a mixture of iron and carbon.
- **Limonite** - A low-grade iron ore with below 50% metallic iron content. It is mostly yellow and contains lime, making it self-fluxing.

World Distribution of Iron Ore and Alloy Metals:

Alloy metals are essential components in the production of high-quality steel. These elements, which include **manganese, molybdenum, nickel, chromium, vanadium, tungsten, and cobalt**, enhance the properties of steel, making it suitable for various industrial applications. **Steels containing at least 1.6% manganese** or any other non-ferrous metal are known as **alloy steels**.

Alloy Metals

A metal alloy is a substance that combines more than one metal or mixes a metal with other non-metallic elements.

Ferrous Metals:

Manganese:

- Manganese is **crucial for producing good-quality steel**, as it decreases brittleness and imparts strength. India possesses significant high-grade manganese ore deposits in **Madhya Pradesh, Odisha, Maharashtra, Goa, Jharkhand, Tamil Nadu, and Karnataka**. The country exports manganese to the USA, the UK, France, and Japan.
 - Globally, **South Africa accounts for about 78% of the world's identified manganese resources**.

Cobalt:

- Cobalt is primarily sourced from **Congo (Kinshasa)**, which **supplies more than half of the world's cobalt production**. This metal is vital for various high-tech and industrial applications.

Molybdenum:

- Molybdenum is extensively used in high-temperature alloys for components in gas **turbines, jet engines, and turbo-superchargers**. It does not occur in nature in a free state, with molybdenite being its principal ore. In India, molybdenum is intermittently produced as a **by-product from the uranium ore of Jadugoda mine in Jharkhand**.
 - **Globally, about two-thirds of molybdenum production is a by-product of copper mining**, with leading producers including **China, Chile, the USA, Peru, and Mexico**.

Tungsten:

- A significant amount of tungsten is recovered through **recycling scrap** tungsten products. India's only operative mines, Degana in Rajasthan and Chendapathar in West Bengal, are now closed.
 - Major global producers of tungsten include **China, Vietnam, Russia, the UK, and Bolivia**.

Vanadium:

- Vanadium's leading producers and reserves are found in **China, Russia, South Africa, and Brazil**. India meets its vanadium demand mainly through imports and some from red mud sludge.

Chromium:

- In India, chromium deposits are found in **Bihar, Karnataka, Maharashtra, Odisha, and Tamil Nadu**.
 - The leading global producers of chromium are **South Africa, Kazakhstan, and India**.

Nickel:

- In India, **Odisha** is the leading producer of nickel, followed by **Karnataka, Kerala, and Rajasthan**.
 - Globally, the leading producers are Indonesia, the **Philippines, New Caledonia, Canada, and Australia**, with **maximum reserves in Australia**, followed by Brazil, Russia, and Cuba.

Non-Ferrous Metals:**Aluminium:**

- Aluminium is a **lightweight and durable metal** primarily **obtained from bauxite**. In India, Odisha is the leading producer, accounting for half of the nation's bauxite production, followed by **Gujarat, Jharkhand, Maharashtra, Chhattisgarh, Tamil Nadu, and Madhya Pradesh**.
 - Globally, **Australia, China, Brazil, Indonesia, Guinea**, and India are the chief producers of bauxite.

Copper:

- Copper occurs naturally and is often **found mixed with gold, silver, lead, and zinc**. Major copper mines in India include the **Khetri** copper belt in **Rajasthan**, the **Singhbhum** copper belt in **Jharkhand**, and the **Malanjkhand** copper belt in **Madhya Pradesh**.
 - Leading copper-producing countries are **Chile, Peru, China, and the USA**.
- **Copper Alloys:**
 - **Brass** - An alloy of copper and zinc.
 - **Bronze** - Typically refers to copper-tin alloys but can include other copper alloys like aluminium bronze.
 - **Carat Silver and Gold Alloys** - Copper is a vital component in these alloys, used in the jewellery industry to modify colour, hardness, and melting point.
 - **Cupronickel** - An alloy of copper and nickel used in low-denomination coins.

Lead:

- In India, the **Zawar mines in Udaipur district of Rajasthan** are the major producers of lead ore.
 - Globally, the leading producers are **China, Australia, the USA, and Peru**, with **Australia holding the maximum reserves**.

Zinc:

- India's zinc production mainly comes from the **Zawar mines in Rajasthan** and the **Arcot districts of Tamil Nadu**.
 - Principal zinc-producing countries are **China, Peru, India, and Australia**, with **Australia having the largest reserves**.

Tin:

- The principal ore of tin is **cassiterite**. In India, **Chhattisgarh** is the only tin-producing state.
 - Globally, **China** has the maximum reserves, followed by **Indonesia, Brazil, Australia, and Bolivia**.

Mica:

- In India, **Gaya-Hazaribagh belt in Jharkhand** accounts for **60% of production**. Nellore in Andhra Pradesh and Rajasthan are other areas of production.
 - Large deposits of mica-bearing rocks are known to exist in countries such as **Brazil, India, and Madagascar**.

Precious Metals:**Silver:**

- In India, the majority of silver production originates from the **Zawar mines of lead in Rajasthan**. Additionally, silver is extracted along with gold from the **Kolar gold fields** and with copper from the **Manbhum region** (Dhanbad and Purulia).
 - Globally, **Peru** holds the maximum reserves of silver, making it a significant player in the silver market.

Platinum

- India's platinum deposits are found primarily in **Orissa** and **Hanumalpur in Karnataka**.
 - **South Africa** stands as the largest producer of platinum worldwide, followed by **Russia, Zimbabwe, Canada, and the USA**. These countries contribute substantially to the global supply of this precious metal.

Gold

- Gold in India is predominantly found in a relatively **pure state**, manifesting as thin veins within rocks. The country has three major goldfields located at **Kolar and Hutti in Karnataka** and **Ramgiri in Andhra Pradesh**.
 - On the global stage, **China, Australia, Russia, and the USA** are leading producers of gold, significantly influencing the international gold market.

Precious Stones:**Diamonds**

- Historically, India was the sole source of diamonds until 1726. Today, major diamond production in India is concentrated in the **Panna mines of Madhya Pradesh**.
 - Internationally, **Russia holds the maximum reserves** of diamonds, followed by **Congo, Australia, and Botswana**, positioning these countries as key players in the diamond industry.

Blood Diamonds

*Blood diamonds, also known as **conflict diamonds**, are mined in areas controlled by rebel forces opposing the government. **These diamonds fund armed conflicts and violent campaigns**. During the 1990s, conflict diamonds were prevalent in **African nations like Congo, Sierra Leone, and Angola**.*

*To combat this, the **Kimberley Process Certification Scheme** was established in **2003** to verify that diamond exports are free from conflict diamonds. India served as the chair and host of the Kimberley Process in 2019.*

Important Non-Metallic Minerals:

Asbestos:

- Asbestos is a group of **naturally occurring fibrous minerals** used for their **heat-resistant properties**. In India, significant deposits of asbestos are found in **Jharkhand, Odisha, Madhya Pradesh, Karnataka, and Andhra Pradesh**. These regions contribute to the country's asbestos production.
 - Globally, the leading producers of asbestos include **Russia, Kazakhstan, China, and Brazil**. The mineral is utilized in various industries due to its durability and resistance to heat and chemicals.

Sulphur:

- Sulphur is predominantly **found in igneous rocks** and is essential for various industrial processes. Approximately **75% of the sulphur mined is used in the production of sulphuric acid**, a crucial chemical in fertilizer manufacture. Additional uses of sulphur include its application in the production of fertilizers and other chemical compounds. In India, major sulphur-producing states include **Chhattisgarh**.
 - On the global stage, **China** is the dominant producer of sulphur, with other significant producers being the **USA, Russia, Canada, and the UAE**.

Gypsum:

- Gypsum is a versatile mineral used primarily in the **construction industry**, particularly in the manufacture of plaster and wallboard. In India, **Rajasthan** holds the majority of gypsum reserves, with additional deposits located in **Jammu and Kashmir, Tamil Nadu**, and other regions.
 - **China** is the leading global producer of gypsum, followed by the **USA, Iran, Thailand, Turkey, and Italy**. The mineral's extensive use in construction makes it a critical resource for building and infrastructure development.

Salt:

- Salt is obtained from various sources, including **seawater, lake water, subsoil water, and rock salt beds**. Rock salt, derived from solid salt beds, is commonly used in the chemical industry and as commercial table salt. In India, **Gujarat is the largest producer of salt**, utilizing methods such as evaporation from seawater and lake water.
 - On a global scale, **China** is the leading producer of salt, followed by **the USA, India, Germany, and Canada**. The mineral's wide range of uses, from culinary to industrial applications, underscores its significance in daily life and industry.

Rocks

Introduction:

The Earth's crust, the outermost layer of the lithosphere, is crucial as it is where human activity occurs. Composed primarily of rocks, the crust varies greatly in hardness, colour, and composition. Rocks are aggregates of minerals, which themselves are made up of two or more elements in a specific ratio.

Types of Rocks - Rocks are categorized based on their formation processes and appearance into **three main types**:

- Igneous Rocks
- Sedimentary Rocks
- Metamorphic Rocks

Igneous Rocks:

The word igneous is derived from the Latin word 'ignis' meaning **fire**. Igneous rocks, also known as **primary rocks**. They are formed from the **solidification of magma** (molten rock below the Earth's surface) **and lava** (molten rock that reaches the surface). As the first rocks on Earth, igneous rocks originate under conditions of high temperatures and are characterized by the **absence of fossils**. **Approximately 95% of the outermost 16 kilometers of Earth's volume consists of igneous rocks**. They are typically hard, massive, and crystalline due to their magmatic origin.

Classification of Igneous Rocks:

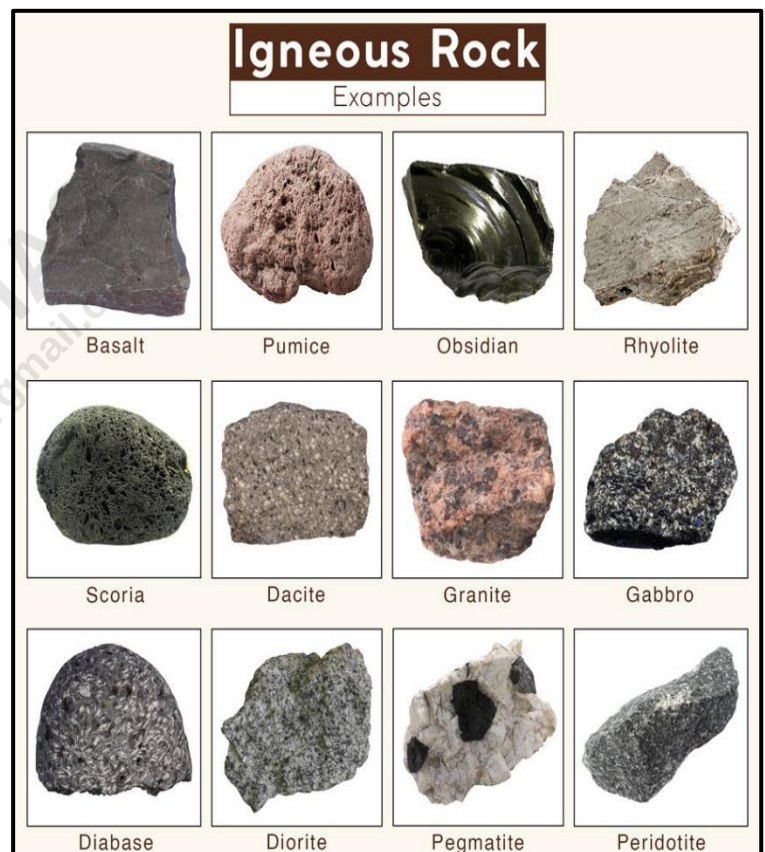
Based on Cooling and Origin:

• Intrusive Igneous Rocks (Plutonic Rocks):

- These rocks form from **magma that cools slowly at great depths** within the Earth. The slow cooling process allows the formation of large mineral grains. Plutonic rocks are **usually medium to large grained** due to their **slow cooling rates**. They appear on the surface only after significant geological processes, such as uplift and denudation, expose them.
- **Examples - Granite, Diorite, and Gabbro** are common plutonic rocks. They are typically exposed at the surface through processes of denudation and erosion.

• Extrusive Igneous Rocks (Volcanic Rocks):

- These rocks form **from lava that cools quickly either just below the surface or at the surface**. Due to their **rapid cooling, volcanic rocks are often fine-grained and can be smooth**. They are typically lighter in colour and less dense compared to plutonic rocks.
- **Examples - Basalt** is a well-known **volcanic rock**, with significant examples including the **Deccan Traps in India** and the **Snake Plateau in Colombia**.



- **Intermediate Rocks (Hypabyssal or Dyke Rocks):**

- These rocks have cooling conditions and characteristics that **fall between those of plutonic and volcanic rocks**. Intermediate rocks display a range of textures and mineral compositions that bridge the gap between intrusive and extrusive rocks.
- **Example - Dyke** rocks are semi-crystalline in structure.

Based on Composition:

- **Acidic Rocks:**

- Acidic rocks are characterized by their **high silica content**, which can reach up to **80%**. This composition primarily includes **quartz and feldspar**. The remaining components consist of aluminium, alkalis, magnesium, iron oxide, and lime.
- Due to **the lower presence of heavier minerals** such as iron and magnesium, acidic rocks are **less dense and lighter in colour** compared to basic rocks.
- These rocks constitute the **sial portion of the Earth's crust**. The high silica content in acidic magma causes it to **cool rapidly, resulting in limited flow and spread**. Consequently, high mountain ranges are often formed from these rocks. Acidic rocks are known for their **hardness, compactness, and resistance to weathering**.
- **Example** - Granite, quartz, and feldspar.

- **Basic Rocks:**

- Basic rocks have a **lower silica content**, around **40% or less**, with a **significant amount of magnesia, up to 40 percent**. The remaining composition includes **iron oxide, lime, aluminium, alkalis, and potassium**.
- Due to their low silica content, the parent material of basic rocks **cools more slowly and flows more extensively, leading to the formation of plateaus**.
- The presence of heavier elements gives these rocks a **dark colour**. Basic rocks are generally **less hard and more susceptible to weathering**.
- **Example** - Basalt, gabbro, and dolerite.

Economic Significance of Igneous Rocks:

- Since magma is the chief source of metal ores, many of them are associated with igneous rocks.
- The minerals of great economic value found in igneous rocks are magnetic iron, nickel, copper, lead, zinc, chromite, manganese, gold, diamond and platinum.
- Amygdales are almond-shaped bubbles formed in basalt due to escape of gases and are filled with minerals.
- The old rocks of the great Indian peninsula are rich in these crystallised minerals or metals.
- Many igneous rocks like granite are used as building materials as they come in beautiful shades.

Sedimentary Rocks:

Sedimentary rocks are formed through the **successive deposition of sediments**, which are fragments derived from previously existing rocks such as igneous, metamorphic, or older sedimentary rocks. These rocks typically exhibit a **layered or stratified structure**, with strata varying in thickness from a few millimeters to several meters, earning them the alternative name of stratified rocks. **Often, fossils—solid parts or impressions of prehistoric plants or animals—are found within these strata.**

Sedimentary rocks cover about **75% of the Earth's crust**, yet they occupy only about **5% of its volume**, primarily because they are found in the upper layers of the crust.

Formation and Classification:

Sedimentary rocks originate from sediments produced by the denudation of various rock types, which includes weathering and erosion. These rocks can be classified based on their mode of formation:

- **Mechanically Formed Sedimentary Rocks:** These rocks are created by mechanical agents such as **running water, wind, ocean currents, and ice**. Examples include:
 - **Arenaceous Rocks** - These rocks have a **high sand content and larger particles**, making them **hard and porous**. They are **excellent reservoirs for groundwater and petroleum**. **Sandstone** is a typical example.
 - **Argillaceous Rocks** - Comprising more clay and finer particles, these rocks are softer and generally less porous. Examples include claystone and shale.
- **Chemically Formed Sedimentary Rocks:** Formed from the **evaporation of water containing minerals**, these rocks can include:
 - **Limestone, Halite, Potash** - These rocks result from chemical precipitation, often observed in environments like **salt lakes and springs**.
- **Organically Formed Sedimentary Rocks:** These rocks are formed from the **remains of plants and animals** that undergo changes due to heat and pressure over time. Examples include:
 - **Coal and Limestone:** Depending on their composition, they can be classified as **either calcareous** (e.g., limestone, chalk) **or carbonaceous** (e.g., coal).

Chief Characteristics:

- **Stratification** - They are layered, consisting of multiple strata.
- **Informative Geological Records** - These rocks preserve detailed geological records due to various physical and biological processes such as weather patterns and fossil formation.
- **Fossiliferous Nature** - They often contain fossils of plants and animals.

- **Porosity** - Generally, they are porous, allowing water to percolate through them.



Spread of Sedimentary Rocks in India:

- **Alluvial Deposits** - Found in the **Indo-Gangetic Plain** and **coastal plains**, these deposits include loam and clay.
- **Sandstones** - Various types of sandstone are present in Madhya Pradesh, eastern Rajasthan, parts of the Himalayas, Andhra Pradesh, Bihar, and Orissa.
- **Vindhyan Highland** - This region in central India is known for its sandstones, shales, and limestones.
- **Coal Deposits** - Significant coal deposits are found in the river basins of the Damodar, Mahanadi, and Godavari, associated with Gondwana sedimentary deposits.

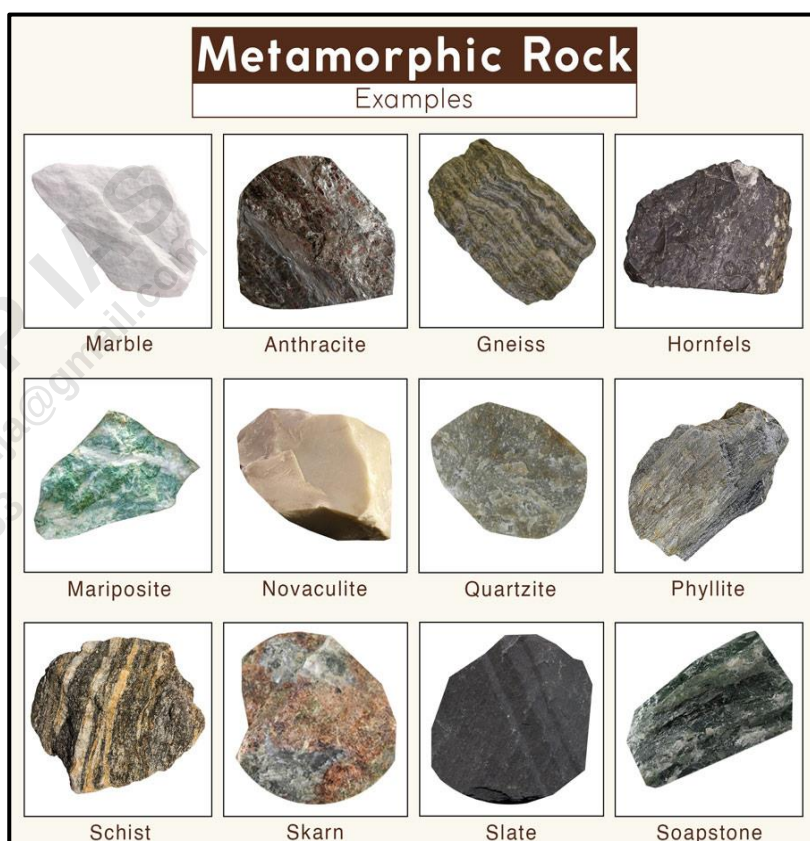
Economic Significance of Sedimentary Rocks:

- Sedimentary rocks are not as rich in minerals of economic value as the igneous rocks.
- But important minerals such as hematite iron ore, phosphates, building stones, coals, petroleum and material used in the cement industry are found.
- The decay of tiny marine organisms yields petroleum. Petroleum occurs in suitable structures only.
- Important minerals like bauxite, manganese, tin, are derived from other rocks but are found in gravels and sands carried by water.
- Sedimentary rocks also yield some of the richest soils.

Metamorphic Rocks:

Metamorphic rocks are formed through the process of **metamorphism**, which involves the **recrystallization and reorganization of minerals within a rock due to changes in pressure, temperature, and volume**. This process occurs when rocks are subjected to significant pressure, either through tectonic movements that force rocks to lower levels, or when molten magma coming through the crust interacts with surrounding crustal rocks.

The term "**metamorphic**" itself means "**change of form**," reflecting the transformation that these rocks undergo during metamorphism. Most rocks in **mountainous regions** show evidence of change.



Characteristics of Metamorphic Rocks:

- During metamorphism, the minerals within the rock may become **aligned in layers or lines**, a feature known as **foliation or lineation**.
- In some cases, minerals or materials of different types are arranged into alternating layers of varying thickness, which is referred to as **banding**.
- Common examples of metamorphic rocks include **gneissoid, slate, schist, marble, and quartzite**.

Causes of Metamorphism:

- **Orogenic (Mountain Building) Movements** - Orogenic movements, associated with **mountain building**, involve **folding, warping**, and **high temperatures** that alter existing rocks. These processes lead to the development of new mineral structures and appearances in the rocks.
- **Lava Inflow** - The intrusion of **molten magmatic material into the earth's crust** can subject surrounding rocks to **intense temperatures and pressures, causing metamorphic changes**. For instance, the peak of **Mount Everest**, composed of metamorphosed **limestone**, illustrates the effects of lava inflow.
- **Geodynamic Forces** - Geodynamic forces, such as plate tectonics, play a crucial role in metamorphism. The **movement and interaction of tectonic plates** contribute to the conditions necessary for metamorphism.

Types of Metamorphism:

- **Thermal Metamorphism:**
 - Thermal metamorphism occurs when rocks experience recrystallization **under high temperatures**. This type of metamorphism is typically associated with magmatic intrusions.
 - **Example - Sandstone can transform into quartzite, and limestone can become marble** as a result of thermal metamorphism. A notable example is the metamorphosed limestone found in the peak of Mount Everest.
- **Dynamic Metamorphism:**
 - Dynamic metamorphism involves the formation of metamorphic rocks **under high pressure**. This can occur with or without accompanying high temperatures and the influence of chemically charged water.
 - When directed pressure and heat are combined, it leads to significant recrystallization and the development of new rock structures. This is known as **dynamo-thermal metamorphism**.
 - **Example - Under high pressure, granite can transform into gneiss, and clay and shale can change into schist.**

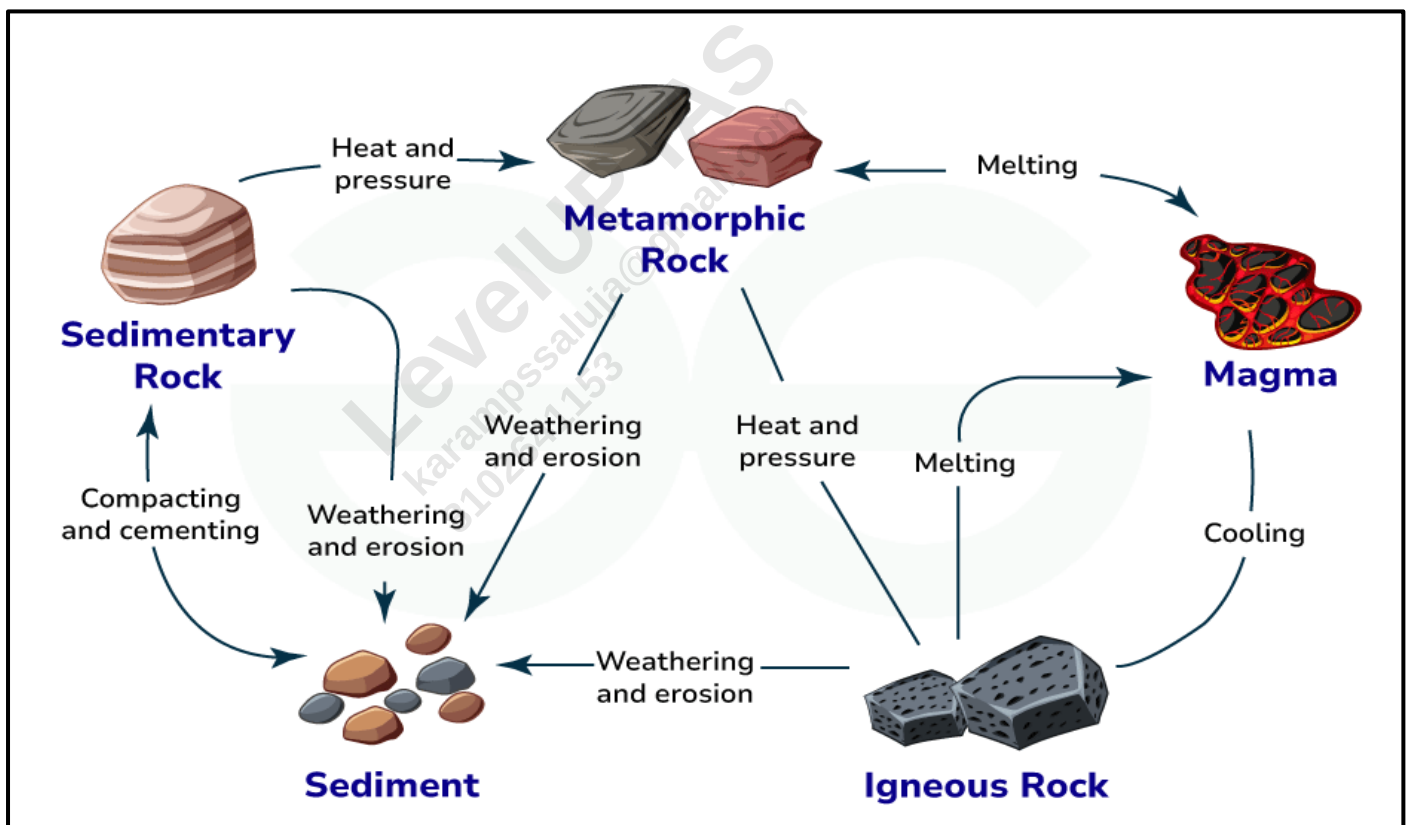
Metamorphic Rocks in India:

- **Gneisses and Schists** - These are prevalent in the Himalayas and extend through Assam, West Bengal, Bihar, Orissa, Madhya Pradesh, and Rajasthan.
- **Quartzite** - This hard rock is commonly found in Rajasthan, Bihar, Madhya Pradesh, Tamil Nadu, and the regions surrounding Delhi.
- **Marble** - Marble deposits are located near Alwar, Ajmer, Jaipur, and Jodhpur in Rajasthan, as well as in parts of the Narmada Valley in Madhya Pradesh.
- **Slate** - Used for roofing and educational writing, slate is found in Rewari (Haryana), Kangra (Himachal Pradesh), and parts of Bihar.
- **Graphite** - This material is present in Orissa and Andhra Pradesh

Rock Cycle:

The rock cycle is a dynamic and **continuous process through which existing rocks are transformed into new types**. This cycle involves the formation, alteration, and recycling of rocks through various geological processes.

- **Igneous Rocks:** These are the primary rocks formed from the cooling and solidification of molten magma. They serve as the **starting point for the rock cycle**.
- **Transformation into Other Rocks:** Igneous rocks can **undergo changes to become sedimentary or metamorphic rocks**. This transformation occurs through **different geological processes**:
 - **Sedimentary Rocks** - Formed from the **accumulation and compaction of fragments derived from the weathering and erosion** of igneous and metamorphic rocks. Over time, these fragments are deposited in **layers and cemented together**.
 - **Metamorphic Rocks** - Created when igneous or sedimentary rocks are **subjected to high pressure and temperature conditions**, causing them to recrystallize and change in mineral composition and structure.
- **Recycling into Magma:** Both sedimentary and metamorphic rocks can be subjected to **high temperatures and pressures**, causing them to melt and **become molten magma**. This molten material then **risers to the surface and cools to form new igneous rocks, completing the cycle**.
- **Subduction Process:** In the Earth's crust, rocks are carried down into the mantle through the process of subduction. As these rocks descend, they **melt and transform into magma**. This magma eventually returns to the surface, contributing to the formation of new igneous rocks.



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